

Online Appendix for: Waste Reduction of Perishable Products through Markdowns at Expiry Dates

EC.1. Proof of Lemma 1

For use in future research, we prove the lemma for general $K \geq 2$, under the assumptions that $\mathbb{P}(M \geq q) > 0$, $p_{0|A} > 0$ for all $A \subset \{1, \dots, K\}$, $p_{1|A} > 0$ for all $A \subset \{1, \dots, K\}$ with $1 \in A$, and $p_{2|A} > 0$ for all $A \subset \{1, \dots, K\}$ with $2 \in A$. Note that this generalizes Assumption 1-(i).

For states $\mathbf{n}, \mathbf{n}'$ in $\mathcal{S}$, we say that there is a path from $\mathbf{n}$ to $\mathbf{n}'$, denoted as $\mathbf{n} \rightarrow \mathbf{n}'$, if there are $l \in \mathbb{N}$ and $\mathbf{n}_{(0)}, \dots, \mathbf{n}_{(l)} \in \mathcal{S}$ such that $\mathbf{n} = \mathbf{n}_{(0)}$, $\mathbf{n}' = \mathbf{n}_{(l)}$, and $P(\mathbf{n}_{(s)}, \mathbf{n}_{(s+1)}) > 0$ for all $s = 0, \dots, l-1$. Write $\mathbf{0} = (0, \dots, 0) \in \mathbb{N}_0^K$, $\mathbf{1} = (1, \dots, 1) \in \mathbb{N}_0^K$, and $\mathbf{e}_k \in \mathbb{N}_0^K$, $k = 1, \dots, K$, for the vector whose k -th component equals one and all other components equal zero. Define the *shift* operator

$$\sigma : \mathcal{S} \rightarrow \mathcal{S}, \quad \sigma((n_1, \dots, n_K)) := (n_2, \dots, n_K, 0), \quad (\text{EC.1})$$

and the *replenishment* operator

$$\rho : \mathcal{S} \rightarrow \mathcal{S}, \quad \rho(\mathbf{n}) := \sigma(\mathbf{n}) + (\mathbf{q} - \sum_{k=2}^K \mathbf{n}_k) \mathbf{e}_K,$$

so that the inventory dynamics satisfy

$$\mathbf{N}(t+1) = \rho(\mathbf{N}(t)) - \mathbf{S}(t+1), \quad (t \in \mathbb{N}_0). \quad (\text{EC.2})$$

Write $\sigma^{(0)}(\mathbf{n}) = \mathbf{n}$, $\sigma^{(j+1)}(\mathbf{n}) = \sigma(\sigma^{(j)}(\mathbf{n}))$ for $j \in \mathbb{N}_0$, and $\sigma^{(-j)}(\mathbf{n}) = \sigma^{(\mathbf{Kj}-\mathbf{j})}(\mathbf{n})$ for $j \in \mathbb{N}$. Note that $\sigma^{(K)}(\mathbf{n}) = \mathbf{n}$, for all $\mathbf{n} \in \mathcal{S}$. To prove irreducibility, we show that $\mathbf{n} \rightarrow \mathbf{n}'$ for any $\mathbf{n}, \mathbf{n}' \in \mathcal{S}$. To this end, we first prove three claims.

Claim 1: Let $\mathbf{n} \in \mathcal{S}$ and $l_1, l_2 \in \mathbb{N}_0$ with $l_1 \leq (\rho(\mathbf{n}))_1$ and $l_2 \leq (\rho(\mathbf{n}))_2$. Then

- (1a) $\mathbb{P}(\mathbf{S}(1) = l_1 \mathbf{e}_1 \mid \rho(\mathbf{N}(0)) = \rho(\mathbf{n})) > 0$;
- (1b) $\mathbb{P}(\mathbf{S}(1) = l_2 \mathbf{e}_2 \mid \rho(\mathbf{N}(0)) = \rho(\mathbf{n})) > 0$;
- (1c) $\mathbb{P}(\mathbf{S}(1) = \rho(\mathbf{n}) \mid \rho(\mathbf{N}(0)) = \rho(\mathbf{n})) > 0$.

Proof of Claim 1. Let $m \geq q$ be such that $\mathbb{P}(M = m) > 0$. Let $\mathbf{n} \in \mathcal{S}$ and l_1, l_2 as in the statement of the claim. On the event $\{M_1 = m\}$ let $X_j \in [K] \cup \{0\}$ denote the purchase of the j -th arriving customer on period 1, for $j = 1, \dots, m$, and let $B_1 := \{j \in [K] : (\rho(\mathbf{n}))_j > 0\}$ and $B_i := \{k \in B_{i-1} : k \neq X_{i-1}\}$ for $i = 2, \dots, m$ denote the corresponding sets of available products. The sales vector $l_1 \mathbf{e}_1$ can be materialized if, for example, precisely m customers arrive of which the first l_1 purchase product 1 and the remaining $m - l_1$ purchase nothing. This occurs with probability

$$\left(\prod_{i=1}^{l_1} p_{1|B_i} \right) \cdot \prod_{i=l_1+1}^m p_{0|B_i} > 0.$$

Similarly, the sales vector $l_2 \mathbf{e}_2$ can be materialized if, for example, precisely m customers arrive of which the first l_2 purchase product 2 and the remaining $m - l_2$ purchase nothing. This occurs with probability

$$\left(\prod_{i=1}^{l_2} p_{2|B_i} \right) \cdot \prod_{i=l_2+1}^m p_{0|B_i} > 0.$$

Finally, the sales vector $\rho(\mathbf{n})$ can be materialized if, for example, precisely m customers arrive of which the first q purchase a product and the remaining $m - q$ purchase nothing. This occurs with probability

$$\left(\prod_{i=1}^q (1 - p_{0|B_i}) \right) \cdot \prod_{i=q+1}^m p_{0|B_i} > 0.$$

This completes the proof of Claim 1. ◇

Claim 2: For all $\mathbf{n} \in \mathcal{S}$, writing $r(\mathbf{n}) := \mathbf{q} - \sum_{k=1}^K \mathbf{n}_k$, the following paths exist:

- (2a) $\mathbf{n} \rightarrow \mathbf{0}$;
- (2b) $\mathbf{n} \rightarrow \rho(\mathbf{n})$;
- (2c) If $r(\mathbf{n}) = \mathbf{0}$ then $\mathbf{n} \rightarrow \sigma(\mathbf{n})$;
- (2d) $\mathbf{n} + \mathbf{r}(\mathbf{n})\mathbf{e}_2 \rightarrow \sigma(\mathbf{n}) + \mathbf{l}\mathbf{e}_1$ for $l = 0, \dots, r(\mathbf{n})$;
- (2e) If $K \geq 3$ then $\mathbf{n} + \mathbf{r}(\mathbf{n})\mathbf{e}_3 \rightarrow \sigma(\mathbf{n}) + \mathbf{l}\mathbf{e}_2$ for $l = 0, \dots, r(\mathbf{n})$;
- (2f) $\mathbf{n} + \mathbf{r}(\mathbf{n})\mathbf{e}_K \rightarrow \sigma^{(-1)}(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_K$.

Proof of Claim 2. For (2a) - (2c) it follows from Claim 1 that

$$\begin{aligned} P(\mathbf{n}, \mathbf{0}) &= \mathbb{P}(\mathbf{S}(\mathbf{1}) = \rho(\mathbf{n}) \mid \rho(\mathbf{N}(\mathbf{0})) = \rho(\mathbf{n})) > \mathbf{0}, \\ P(\mathbf{n}, \rho(\mathbf{n})) &= \mathbb{P}(\mathbf{S}(\mathbf{1}) = \mathbf{0} \mid \rho(\mathbf{N}(\mathbf{0})) = \rho(\mathbf{n})) > \mathbf{0}, \\ P(\mathbf{n}, \sigma(\mathbf{n})) &= \mathbb{P}(\mathbf{S}(\mathbf{1}) = \mathbf{0} \mid \rho(\mathbf{N}(\mathbf{0})) = \rho(\mathbf{n})) > \mathbf{0} \quad \text{if } \mathbf{r}(\mathbf{n}) = \mathbf{0}, \end{aligned}$$

where the last line uses $\rho(\mathbf{n}) = \sigma(\mathbf{n})$ if $r(\mathbf{n}) = \mathbf{0}$. For (2d) - (2e), note that $r(\mathbf{n}) > \mathbf{0}$ and $l \in \{0, \dots, r(\mathbf{n})\}$ implies

$$\begin{aligned} P(\mathbf{n} + \mathbf{r}(\mathbf{n})\mathbf{e}_2, \sigma(\mathbf{n}) + \mathbf{l}\mathbf{e}_1) &= \mathbb{P}(\mathbf{S}(\mathbf{1}) = (\mathbf{r}(\mathbf{n}) - \mathbf{l})\mathbf{e}_1 \mid \rho(\mathbf{N}(\mathbf{0})) = \sigma(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_1), \\ P(\mathbf{n} + \mathbf{r}(\mathbf{n})\mathbf{e}_3, \sigma(\mathbf{n}) + \mathbf{l}\mathbf{e}_2) &= \mathbb{P}(\mathbf{S}(\mathbf{1}) = (\mathbf{r}(\mathbf{n}) - \mathbf{l})\mathbf{e}_2 \mid \rho(\mathbf{N}(\mathbf{0})) = \sigma(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_2) \text{ if } K \geq 3, \end{aligned}$$

and apply Claim 1. For $r(\mathbf{n}) = \mathbf{0}$, (2d) and (2e) follow from (2c). For (2f), if $r(\mathbf{n}) > \mathbf{0}$ and $K \geq 3$ then

$$\begin{aligned} \mathbf{n} + \mathbf{r}(\mathbf{n})\mathbf{e}_K &\xrightarrow{(2c)} \sigma^{(K-3)}(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_3 = \sigma^{(\mathbf{K}-3)}(\mathbf{n}) + \mathbf{r}(\sigma^{(\mathbf{K}-3)}(\mathbf{n}))\mathbf{e}_3 \xrightarrow{(2e)} \sigma^{(\mathbf{K}-2)}(\mathbf{n}) \\ &\xrightarrow{(2b)} \sigma^{(K-1)}(\mathbf{n}) + \mathbf{r}(\sigma^{(\mathbf{K}-1)}(\mathbf{n}))\mathbf{e}_K = \sigma^{(-1)}(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_K, \end{aligned}$$

and if $r(\mathbf{n}) > \mathbf{0}$ and $K = 2$ then, by Claim 1,

$$P(\sigma(\mathbf{n}) + \mathbf{r}\mathbf{e}_1, \mathbf{n}) = \mathbb{P}(\mathbf{S}(\mathbf{1}) = \mathbf{r}\mathbf{e}_2 \mid \rho(\mathbf{N}(\mathbf{0})) = \mathbf{n} + \mathbf{r}\mathbf{e}_2) > \mathbf{0},$$

and therefore

$$\mathbf{n} + \mathbf{r}(\mathbf{n})\mathbf{e}_2 \xrightarrow{(2c)} \sigma(\mathbf{n} + \mathbf{r}\mathbf{e}_2) = \sigma(\mathbf{n}) + \mathbf{r}\mathbf{e}_1 \rightarrow \mathbf{n} \xrightarrow{(2b)} \rho(\mathbf{n}) = \sigma^{-1}(\mathbf{n}) + \mathbf{r}\mathbf{e}_2.$$

For $r(\mathbf{n}) = 0$, (2f) follows from (2c). This completes the proof of Claim 2. $\diamond$

Claim 3: Let $\mathbf{n} \in \mathcal{S}$ with $\sum_{k=1}^K n_k < q$. Then $\mathbf{n} \rightarrow \mathbf{n} + \mathbf{e}_i$ for all $i = 1, \dots, K$.

Proof of Claim 3. Let $\mathbf{n} \in \mathcal{S}$ such that $r(\mathbf{n}) := \mathbf{q} - \sum_{k=1}^K \mathbf{n}_k > \mathbf{0}$. If $i = 1$ then

$$\mathbf{n} \xrightarrow{(2b)} \rho(\mathbf{n}) = \sigma(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_K \xrightarrow{(2c)} \sigma^{(1+K-2)}(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_2 \xrightarrow{(2d)} \sigma^{(1+K-2+1)}(\mathbf{n}) + \mathbf{e}_1 = \mathbf{n} + \mathbf{e}_1.$$

Suppose $i \in \{2, \dots, K\}$. If $K = 2$ then Claim 1 implies

$$P(\mathbf{n}, \sigma(\mathbf{n})) = \mathbb{P}(\mathbf{S}(1) = \mathbf{r}(\mathbf{n})\mathbf{e}_2 \mid \rho(\mathbf{N}(0)) = \sigma(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_2) > 0,$$

$$P(\sigma(\mathbf{n}), \sigma^{(2)}(\mathbf{n}) + \mathbf{e}_2) = \mathbb{P}(\mathbf{S}(1) = (\mathbf{r}(\mathbf{n}) - 1)\mathbf{e}_2 \mid \rho(\mathbf{N}(0)) = \sigma^{(2)}(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_2) > 0,$$

from which it follows that

$$\mathbf{n} \rightarrow \sigma(\mathbf{n}) \rightarrow \sigma^{(2)}(\mathbf{n}) + \mathbf{e}_2 = \mathbf{n} + \mathbf{e}_2 = \mathbf{n} + \mathbf{e}_i.$$

If $K \geq 3$ then

$$\begin{aligned} \mathbf{n} &\xrightarrow{(2b)} \sigma(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_K \xrightarrow{(2f)} \sigma^{(i)}(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_K \xrightarrow{(2c)} \sigma^{(i+K-3)}(\mathbf{n}) + \mathbf{r}(\mathbf{n})\mathbf{e}_3 \xrightarrow{(2e)} \sigma^{(i+K-2)}(\mathbf{n}) + \mathbf{e}_2 \\ &\xrightarrow{(2b)} \sigma^{(i+K-1)}(\mathbf{n}) + \mathbf{e}_1 + (\mathbf{r}(\mathbf{n}) - 1)\mathbf{e}_K \xrightarrow{(2f)} \sigma^{(i+K-1+3-i)}(\mathbf{n}) + \sigma^{(3-i)}(\mathbf{e}_1) + (\mathbf{r}(\mathbf{n}) - 1)\mathbf{e}_K \\ &\xrightarrow{(2c)} \sigma^{(i+K-1+3-i+K-3)}(\mathbf{n}) + \sigma^{(3-i+K-3)}(\mathbf{e}_1) + (\mathbf{r}(\mathbf{n}) - 1)\mathbf{e}_3 \\ &\xrightarrow{(2e, l=0)} \sigma^{(i+K-1+3-i+K-3+1)}(\mathbf{n}) + \sigma^{(3-i+K-2)}(\mathbf{e}_1) = \mathbf{n} + \mathbf{e}_i. \end{aligned}$$

This completes the proof of Claim 3. $\diamond$

By Claim 2a and repeated application of Claim 3, we have $\mathbf{n} \rightarrow \mathbf{0}$, $\mathbf{0} \rightarrow \mathbf{n}'$, and thus $\mathbf{n} \rightarrow \mathbf{n}'$, for all $\mathbf{n}, \mathbf{n}' \in \mathcal{S}$. This proves irreducibility. Aperiodicity follows from $P(\mathbf{0}, \mathbf{0}) > 0$. Because $(\mathbf{N}(t) : t \in \mathbb{N}_0)$ is a Markov chain on a finite state space, Billingsley (1995, Example 8.14) implies the existence of a stationary distribution. Uniqueness of and convergence to the stationary distribution follows from Billingsley (1995, Theorem 8.6). This completes the proof of the lemma.

EC.2. Proof of Theorem 2

In this section we prove the convergence results underpinning Theorem 2. As Theorem 2 is written in an succinct way, we re-write Theorem 2 by expanding the expressions as in the following lemma. Lemma EC.1 identifies three limiting regimes defined by the magnitude of q compared to other parameters: the *abundant inventory regime*, where $q \geq 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$, the *moderate inventory regime* where $\lambda p_{2|2} < q < 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$, and the *scarce inventory regime*, where $q \leq \lambda p_{2|2}$. For each of these three regimes we state the limiting scaled performance metrics. Theorem 2 follows directly from Lemma EC.1.

LEMMA EC.1. *Suppose $p_{1|1,2} + p_{2|1,2} > p_{2|2}$ and $p_{2|2} > p_{2|1,2}$.*

1. *If $q \geq 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$, then*

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{\text{WASTE}^{(m)}}{m} &= \frac{q - 2\lambda p_{1|1,2} - \lambda p_{2|1,2}}{2}, \\ \lim_{m \rightarrow \infty} \frac{\text{REVENUE}^{(m)}}{m} &= r_1 \lambda p_{1|1,2} + r_2 \lambda p_{2|1,2}, \\ \lim_{m \rightarrow \infty} \frac{\text{REPLENISHMENT}^{(m)}}{m} &= \frac{q + \lambda p_{2|1,2}}{2}, \\ \lim_{m \rightarrow \infty} \frac{\text{PROFIT}^{(m)}}{m} &= (r_1 - s) \lambda p_{1|1,2} + \left(r_2 - \frac{c+s}{2} \right) \lambda p_{2|1,2} + \frac{(s-c)q}{2}. \end{aligned}$$

2. *If $\lambda p_{2|2} < q < 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$, then*

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{\text{WASTE}^{(m)}}{m} &= 0, \\ \lim_{m \rightarrow \infty} \frac{\text{REVENUE}^{(m)}}{m} &= r_1 \frac{(q - \lambda p_{2|2})p_{1|1,2}}{2p_{1|1,2} + p_{2|1,2} - p_{2|2}} + r_2 \left(q - 2 \frac{(q - \lambda p_{2|2})p_{1|1,2}}{2p_{1|1,2} + p_{2|1,2} - p_{2|2}} \right), \\ \lim_{m \rightarrow \infty} \frac{\text{REPLENISHMENT}^{(m)}}{m} &= q - \frac{(q - \lambda p_{2|2})p_{1|1,2}}{2p_{1|1,2} + p_{2|1,2} - p_{2|2}}, \\ \lim_{m \rightarrow \infty} \frac{\text{PROFIT}^{(m)}}{m} &= (r_1 - c) \frac{(q - \lambda p_{2|2})p_{1|1,2}}{2p_{1|1,2} + p_{2|1,2} - p_{2|2}} + (r_2 - c) \left(q - 2 \frac{(q - \lambda p_{2|2})p_{1|1,2}}{2p_{1|1,2} + p_{2|1,2} - p_{2|2}} \right). \end{aligned}$$

3. *If $0 \leq q \leq \lambda p_{2|2}$, then*

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{\text{WASTE}^{(m)}}{m} &= 0, \\ \lim_{m \rightarrow \infty} \frac{\text{REVENUE}^{(m)}}{m} &= r_2 q, \\ \lim_{m \rightarrow \infty} \frac{\text{REPLENISHMENT}^{(m)}}{m} &= q, \\ \lim_{m \rightarrow \infty} \frac{\text{PROFIT}^{(m)}}{m} &= (r_2 - c)q. \end{aligned}$$

EC.2.1. Outline of the proof of Lemma EC.1

To establish Lemma EC.1, we have to prove convergence of the performance metrics to the constants provided in the theorem. The proof is based on convergence of the stationary distribution of the Markov chain $(\frac{1}{m}\mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$ to a constant under the mean-field scaling. We demonstrate this convergence in two steps in the moderate inventory regime. Convergence of the performance metrics in the other regimes then follows from a comparison.

In the first step, we connect the behavior of the Markov chain $(\frac{1}{m}\mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$ (describing the inventory at the end of the period) to the behavior of the sales. Assuming that the initial condition of the Markov chain converges weakly to a constant, we exploit this connection to show that the Markov chain

converges weakly to a deterministic dynamical system uniformly on finite time intervals under the mean-field scaling, regardless of any assumptions on the purchase probabilities. The convergence is proven using counting processes and their inverses.

In the second step, we analyze the fixed point of the dynamical system to derive convergence of the stationary distribution. Roughly speaking, if the dynamical system has a globally attracting fixed point, then the stationary distribution converges to this fixed point. We prove that the dynamical system found in the first step indeed has a globally attracting fixed point if the purchasing probabilities satisfy the intuitive conditions stated in Lemma EC.1 in the moderate inventory regime. This proof is based on a careful analysis of the map ϕ characterizing the dynamical system. Even though ϕ has a Lipschitz constant of at least 1, we show that it has a globally attracting fixed point under the aforementioned conditions.

In the final step, we conclude that the stationary inventory vector $\frac{1}{m}\mathbf{N}^{(m)}$ converges weakly to the fixed point in the moderate inventory regime. Then also the performance metrics converge to constants, which are relatively easily expressed in terms of the model parameters. We extend these results for the performance metrics to the other two regimes using comparison arguments.

EC.2.2. Proof of Lemma EC.1

We start with the weak convergence of $(\frac{1}{m}\mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$. To establish convergence of $(\frac{1}{m}\mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$ we show that, if the scaled inventory $\frac{1}{m}\mathbf{N}^{(m)}(t)$ at the end of period t converges weakly to a constant vector (r', r) , then the scaled inventory $(\frac{1}{m}\mathbf{N}_2^{(m)}(t), \frac{1}{m}q^{(m)} - \frac{1}{m}\mathbf{N}_2^{(m)}(t))$ at the start of period $t+1$ converges weakly to $(r, q-r)$. The next lemma shows that, under this condition, the scaled sales vector converges weakly to a constant. Consequently, $\frac{1}{m}\mathbf{N}^{(m)}(t+1)$ converges weakly to a constant, too. This gives rise to a dynamical system whose behavior is demonstrated in Table EC.1 following the lemma.

LEMMA EC.2. *Let $r \in [0, q]$. Define*

$$\begin{aligned} \bar{\chi}(r) &= \lambda \wedge \frac{r}{p_{1|1,2}} \wedge \frac{q-r}{p_{2|1,2}}, \\ s_1(r) &= p_{1|1,2}\bar{\chi}(r) + p_{1|1}(\lambda - \bar{\chi}(r)) \wedge (r - p_{1|1,2}\bar{\chi}(r)), \end{aligned} \quad (\text{EC.3})$$

$$s_2(r) = p_{2|1,2}\bar{\chi}(r) + p_{2|2}(\lambda - \bar{\chi}(r)) \wedge ((q-r) - p_{2|1,2}\bar{\chi}(r)). \quad (\text{EC.4})$$

Let $t \in \mathbb{N}_0$. Assume that the inventory at the end of period t satisfies $\frac{1}{m}\mathbf{N}^{(m)}(t) \Rightarrow (r', r)$, for some $r' \in [0, q]$, and that the number of customers arriving on period $t+1$ scales according to $\frac{1}{m}M_{t+1}^{(m)} \Rightarrow \lambda$. Then the inventory at the start of period $t+1$ satisfies

$$\left(\frac{1}{m}\mathbf{N}_2^{(m)}(t), \frac{1}{m}q^{(m)} - \frac{1}{m}\mathbf{N}_2^{(m)}(t) \right) \Rightarrow (r, q-r),$$

while the sales on that period admit limits

$$\frac{1}{m} \text{SALES}_1^{(m)}(t+1) \Rightarrow s_1(r) \quad \text{and} \quad \frac{1}{m} \text{SALES}_2^{(m)}(t+1) \Rightarrow s_2(r).$$

Consequently, the inventory at the end of period $t+1$ satisfies

$$\frac{1}{m} \mathbf{N}^{(m)}(t+1) \Rightarrow (r - s_1(r), q - r - s_2(r)).$$

Lemma EC.2 demonstrates that if the inventory $\frac{1}{m} \mathbf{N}^{(m)}(t)$ at the end of period t converges weakly to (r', r) , then the inventory $\frac{1}{m} \mathbf{N}^{(m)}(t+1)$ at the end of period $t+1$ converges weakly to $(r - s_1(r), q - r - s_2(r))$. Consequently, if the initial condition $\mathbf{N}^{(m)}(0)$ of the scaled irreducible Markov chain $(\frac{1}{m} \mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$ converges weakly to a constant, then the Markov chain converges weakly to the deterministic dynamical system characterized in Table EC.1.

Table EC.1 **Limiting scaled inventory dynamics.**

prod.	end period t	begin period $t+1$	end period $t+1$
1	r'	r	$r - s_1(r)$
2	r	$q - r$	$q - r - s_2(r)$

We now turn to the weak convergence of the stationary inventory vector $\frac{1}{m} \mathbf{N}^{(m)}$. We can use the previous result to study this: if we can show that the dynamical system has a globally attracting fixed point, then the results of Benaïm and Le Boudec (2011) ensure that the stationary distribution converges weakly to that fixed point. A quick inspection of the dynamics illustrated in Table EC.1 reveals that it suffices to prove that the map

$$\phi(r) := q - r - s_2(r), \quad r \in [0, q],$$

has a globally attracting fixed point. The next lemma shows that this holds under the conditions formulated in Lemma EC.1 in the moderate inventory regime.

LEMMA EC.3. *Suppose that $p_{1|1,2} + p_{2|1,2} > p_{2|2}$ and $p_{2|2} > p_{2|1,2}$. Then ϕ has a globally attracting fixed point*

$$r^* := \frac{(q - p_{2|2})p_{1|1,2}}{2p_{1|1,2} + p_{2|1,2} - p_{2|2}} \tag{EC.5}$$

in the moderate inventory regime.

The previous lemma, combined with a comparison argument, implies the following convergence result.

LEMMA EC.4. *Suppose that $p_{1|1,2} + p_{2|1,2} > p_{2|2}$ and $p_{2|2} > p_{2|1,2}$. Then the expectation of the scaled inventory vector in stationarity $\frac{1}{m}\mathbf{N}^{(m)}$ converges to $(\bar{r} - s_1(\bar{r}), \bar{r})$, where*

$$\bar{r} := \begin{cases} 0 & \text{if } q \leq \lambda p_{2|2}; \\ r^* & \text{if } \lambda p_{2|2} < q < 2\lambda p_{1|1,2} + \lambda p_{2|1,2}; \\ \frac{q - \lambda p_{2|1,2}}{2} & \text{if } q \geq p_{1|1,2} + \lambda p_{2|1,2}. \end{cases} \quad (\text{EC.6})$$

Since the expectation of the scaled inventory vector in stationarity $\frac{1}{m}\mathbf{N}^{(m)}$ converges to $(\bar{r} - s_1(\bar{r}), \bar{r})$, the scaled performance metrics converge to constants, too. The corresponding expressions follow directly from the explicit expressions for $\bar{r}$ and r^* combined with the functions s_1 and s_2 for the sales. This proves Lemma EC.1.

EC.2.3. Proof of Lemma EC.2

To keep notation manageable, we focus on the case with deterministic initial conditions of the form $\lfloor mr \rfloor$ items of product 1 and $\lfloor mq \rfloor - \lfloor mr \rfloor$ items of product 2 at the start of the period in the m -th replica system. Stochastic initial conditions can be treated in a similar way. We say that a given customer experiences scenario j with $j \in \{0, 1, 2, 3\}$, where each scenario is characterized by which products are still for sale.

- Scenario 3: products 1 and 2 available.
- Scenario 2: only product 2 available.
- Scenario 1: only product 1 available.
- Scenario 0: no products available.

The customer pool on a given period can be regarded as a potentially infinite line of customers that enter the store one by one. It is convenient to explicitly model the purchase of a customer for each of the four possible scenarios. For the k -th customer we introduce the random variables

$$A_{(3),k}, A_{(2),k}, A_{(1),k}, \text{ and } A_{(0),k}$$

describing the consumer's purchase. More precisely, $A_{(j),k}$ takes values in the set of products $\{0, a, b\}$ and describes the purchase of customer k given that scenario j is observed. These random variables do not depend on the scaling parameter m : this parameter only influences how many customers enter the store and how many products are in stock. For the moment, there is no need to put any assumption on the distribution of the $A_{(j),k}$, so we postpone that until needed.

We define the random variable

$$X_{(i,j),k} = \mathbf{1}_{\{A_{(j),k}=i\}},$$

which equals 1 if customer k observes scenario j and buys product i . In addition, we define the counting processes $S_{(i,j)}$ and $S_{(i,j),n}$ with $n \in \mathbb{N}$ via

$$S_{(i,j)}(t) = \sum_{k=1}^{\lfloor t \rfloor} X_{(i,j),k}$$

and

$$S_{(i,j),n}(t) = S_{(i,j)}((n-1) + t) - S_{(i,j)}(n-1).$$

The random variable $S_{(i,j)}(t)$ denotes how many items of product i are sold if scenario j is observed by the first t customers. The interpretation of $S_{(i,j),n}(t)$ is slightly more complicated: it denotes how many items of product i are sold if scenario j is observed by the first t customers arriving after the $(n-1)$ -th customer.

The crucial element in this discussion is the use of the inverse of $S_{(i,j)}$. Recall that there are $\lfloor mr \rfloor$ items of product 1 and $\lfloor mq \rfloor - \lfloor mr \rfloor$ items of product 2 at the start of the period. The random number of customers showing up is denoted by $M^{(m)}$. Then we define

$$\tau_{(1,3)}^{(m)} = S_{(1,3)}^{-1}(\lfloor mr \rfloor - 1)$$

and

$$\tau_{(2,3)}^{(m)} = S_{(2,3)}^{-1}(\lfloor mq \rfloor - \lfloor mr \rfloor - 1)$$

We may interpret $\tau_{(1,3)}^{(m)}$ as the number of customers that have to arrive in scenario 3 in order to sell all items of product 1 during that scenario. A similar interpretation holds for $\tau_{(2,3)}$.

Now observe the following. Initially, both product 1 and product 2 are available, so the first customers observe scenario 3, until one of the products is sold out. We know how many customers observe scenario 3, namely

$$\chi_3^{(m)} = M^{(m)} \wedge \tau_{(1,3)} \wedge \tau_{(2,3)}.$$

Consequently, we sell

$$S_{(1,3)}\left(\chi_3^{(m)}\right)$$

items of product 1 to customers that experience scenario 3, and we sell

$$S_{(2,3)}\left(\chi_3^{(m)}\right)$$

items of product 2 to customers that experience scenario 3.

At the end of scenario 3, there are three possible outcomes: there are no more customers left, product 1 is sold out, or product 2 is sold out. What happens next is determined by which outcome occurs. We incorporate this in how we deal with the scenarios 2 and 1.

To make things concrete, consider scenario 2. The number of customers left is $M^{(m)} - \chi_3^{(m)}$, which may be equal to 0. When scenario 3 ends, we have $\lfloor qm \rfloor - \lfloor mr \rfloor - S_{(b,3)}(\chi_3^{(m)})$ items of product 2 left. Then we need

$$\tau_{(2,2)}^{(m)} = \left(S_{(2,2),\chi_3^{(m)}+1} \right)^{-1} \left(\lfloor qm \rfloor - \lfloor mr \rfloor - S_{(2,3)}(\chi_3^{(m)}) - 1 \right)$$

customers to sell all these items of product 2. (We use $S_{(2,2),\chi_3^{(m)}+1}$ because the first $\chi_3^{(m)}$ customers have already arrived.) Consequently, we sell

$$S_{(2,2),\chi_3^{(m)}+1}(\chi_2^{(m)})$$

items of product 2 in scenario 2, where

$$\chi_2^{(m)} = \left[M^{(m)} - \chi_3^{(m)} \right] \wedge \tau_{(2,2)}^{(m)}$$

is the total number of customers that observe scenario 2. For similar reasons, we sell

$$S_{(1,1),\chi_3^{(m)}+1}(\chi_1^{(m)})$$

items of product 1 in scenario 1, where

$$\chi_1^{(m)} = \left[M^{(m)} - \chi_3^{(m)} \right] \wedge \tau_{(1,1)}^{(m)}$$

and

$$\tau_{(1,1)}^{(m)} = \left(S_{(1,1),\chi_3^{(m)}+1} \right)^{-1} \left(\lfloor mr \rfloor - S_{(1,3)}(\chi_3^{(m)}) - 1 \right).$$

So far, we have only translated the number of sold items in terms of counting processes and their inverses. We now proceed to study the asymptotic regime of Lemma EC.2 under the assumption that the buying behavior of different customers is independent. More formally, we assume that $A_{(j),1}, A_{(j),2}, \dots$ forms a collection of i.i.d. random variables for each fixed scenario j . It is also convenient to denote

$$p_{(i,j)} = \mathbb{P} \left(\mathbb{1}_{\{A_{(j),1}=i\}} = 1 \right),$$

so $p_{(i,j)}$ is the probability that a customer purchases product i when facing scenario j . (This means that $p_{(1,3)} = p_{1|1,2}$, for instance. The new notation for purchasing probabilities enables us to write down limits in a more compact way.)

Recall that we are interested in what happens when m grows large. We assume that $\frac{1}{m}M^{(m)} \Rightarrow \lambda$. We also define $S_{i,j}^{(m)}$ via $S_{i,j}^{(m)}(t) = S_{i,j}^{(m)}(mt)$. Given our distributional assumptions, Whitt (2002, Theorem 7.3.2) implies that

$$\frac{1}{m}S_{(i,j)}^{(m)} \Rightarrow \eta_{p(i,j)}$$

and

$$\frac{1}{m} \left(S_{(i,j)}^{(m)} \right)^{-1} \Rightarrow \eta_{1/p(i,j)}$$

as $m \rightarrow \infty$, where $\eta_p(t) = pt$.

Using these results together with the Continuous Mapping Theorem, we may derive convergence of all other quantities of interest. For instance, we get the convergence

$$\frac{1}{m}\tau_{(1,3)}^{(m)} \Rightarrow r/p_{1|1,2} =: \bar{\tau}_{(1,3)}.$$

Similarly, we get

$$\frac{1}{m}\tau_{(2,3)}^{(m)} = \frac{1}{m}S_{(2,3)}^{-1} \left(m \frac{1}{m} ([mq] - [mr] - 1) \right) \Rightarrow (q-r)/p_{2|1,2} =: \bar{\tau}_{(2,3)},$$

so

$$\frac{1}{m}\chi_3^{(m)} = \frac{1}{m}M^{(m)} \wedge \frac{1}{m}\tau_{(1,3)} \wedge \frac{1}{m}\tau_{(2,3)} \Rightarrow \lambda \wedge r/p_{1|1,2} \wedge (q-r)/p_{2|1,2} := \bar{\chi}.$$

Combining this with the earlier obtained convergence results, it follows that

$$\frac{1}{m}S_{(1,3)} \left(\chi_3^{(m)} \right) = \frac{1}{m}S_{(1,3)} \left(m \frac{1}{m}\chi_3^{(m)} \right) \Rightarrow \bar{\sigma}_{(1,3)}$$

with

$$\bar{\sigma}_{(1,3)} := \eta_{p_{1|1,2}}(\bar{\chi}) = p_{1|1,2}\lambda \wedge \frac{p_{1|1,2}}{p_{2|1,2}}(q-r) \wedge r,$$

and

$$\frac{1}{m}S_{(2,3)} \left(\chi_3^{(m)} \right) = \frac{1}{m}S_{(2,3)} \left(m \frac{1}{m}\chi_3^{(m)} \right) \Rightarrow \bar{\sigma}_{(2,3)}$$

with

$$\bar{\sigma}_{(2,3)} := \eta_{p_{2|1,2}}(\bar{\chi}) = p_{2|1,2}\lambda \wedge (q-r) \wedge \frac{p_{2|1,2}}{p_{1|1,2}}r.$$

With these convergence results at our disposal, we can also derive the asymptotics in scenarios 2 and 1. As a first step, we define $S_{(i,j),n}^{(m)}$ via $S_{(i,j),n}^{(m)}(t) = S_{(i,j),n}(mt)$. Suppose that $\frac{1}{m}n^{(m)} \Rightarrow x$ for a finite constant x . Because

$$\frac{1}{m}S_{(i,j),n^{(m)}}^{(m)}(t) = \frac{1}{m}S_{(i,j)}^{(m)}\left(m\frac{1}{m}(n^{(m)} - 1 + mt)\right) - \frac{1}{m}S_{(i,j)}^{(m)}\left(m\frac{1}{m}(n^{(m)} - 1)\right),$$

the earlier obtained asymptotics imply that

$$\frac{1}{m}S_{(i,j),n^{(m)}}^{(m)} \Rightarrow \eta_{p(i,j)}.$$

Now, for scenario 2, observe that

$$\begin{aligned} \frac{1}{m}\tau_{(2,2)}^{(m)} &= \frac{1}{m} \left(S_{(2,2),\chi_3^{(m)}+1} \right)^{-1} \left(\lfloor qm \rfloor - \lfloor mr \rfloor - S_{(2,3)} \left(\chi_3^{(m)} \right) - 1 \right) \\ &\Rightarrow \eta_{1/p_{2|2}} \left((q-r) - \bar{\sigma}_{(2,3)} \right) = \left((q-r) - \bar{\sigma}_{(2,3)} \right) / p_{2|2} \end{aligned}$$

and hence

$$\frac{1}{m}\chi_2^{(m)} = \frac{1}{m} \left[M^{(m)} - \chi_3^{(m)} \right] \wedge \frac{1}{m}\tau_{(2,2)}^{(m)} \Rightarrow [\lambda - \bar{\chi}] \wedge \left((q-r) - \bar{\sigma}_{(2,3)} \right) / p_{2|2}.$$

Then it follows that

$$\frac{1}{m}S_{(2,2),\chi_3^{(m)}+1} \left(\chi_2^{(m)} \right) \Rightarrow p_{2|2} [\lambda - \bar{\chi}] \wedge \left((q-r) - \bar{\sigma}_{(2,3)} \right).$$

In the same way, we obtain that

$$\frac{1}{m}S_{(1,1),\chi_3^{(m)}+1} \left(\chi_1^{(m)} \right) \Rightarrow p_{1|1} [\lambda - \bar{\chi}] \wedge \left(r - \bar{\sigma}_{(1,3)} \right).$$

Combining all convergence results establishes the validity of Lemma EC.2.

EC.2.4. Proof of Lemma EC.3

The two-dimensional system state $(\frac{1}{m}\mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$ constitutes an irreducible Markov chain taking values in the space $[0, 2q]^2$ for all $m \in \mathbb{N}$. In view of the results of Benaïm and Le Boudec (2011), it suffices to show weak convergence of the Markov chain to a deterministic semi-flow on bounded intervals and verify that the latter has a globally attracting fixed point.

We focus on the moderate-inventory regime and assume that $\lambda p_{2|2} < q < 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$. Lemma EC.2 implies that the Markov chain indeed converges to a deterministic semi-flow whose dynamics are illustrated in Table EC.1. Thus it remains to prove that this semi-flow has a globally attracting fixed point. To prove this result it suffices to show that the map $\phi(r) = q - r - s_2(r)$ has a globally attracting fixed point r^* . Then the limiting dynamical system of $(\frac{1}{m}\mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$ has a globally attracting fixed point $(r^* - s_1(r^*), r^*)$ and

the results of Benaim and Le Boudec (2011) imply that the scaled inventory vector in stationarity $\frac{1}{m}\mathbf{N}^{(m)}$ converges weakly to fixed point $(r^* - s_1(r^*), r^*)$.

For the fixed-point analysis of the map ϕ , we rewrite ϕ in a more convenient form. Using the expression for $s_2(r)$, we get

$$\begin{aligned}\phi(r) &= q - r - p_{2|1,2}\bar{\chi}(r) - (p_{2|2}(\lambda - \bar{\chi}(r)) \wedge (q - r - p_{2|1,2}\bar{\chi}(r))) \\ &= (q - r + (p_{2|2} - p_{2|1,2})\bar{\chi}(r) - p_{2|2}\lambda) \vee 0.\end{aligned}$$

Define

$$\bar{p} := (p_{2|2} - p_{2|1,2})/p_{1|1,2} \in (0, 1).$$

The term $\bar{\chi}(r)$ is given as the minimum of three other terms. However, if $\bar{\chi}(r) = (q - r)/p_{2|1,2}$, then

$$q - r + (p_{2|2} - p_{2|1,2})\bar{\chi}(r) - p_{2|2}\lambda = p_{2|2}(\bar{\chi}(r) - \lambda) \leq 0,$$

so

$$\begin{aligned}\phi(r) &= \left(q - r + (p_{2|2} - p_{2|1,2}) \left(\lambda \wedge \frac{r}{p_{1|1,2}} \right) - p_{2|2}\lambda \right) \vee 0 \\ &= (q - r + \bar{p}(r \wedge p_{1|1,2}\lambda) - p_{2|2}\lambda) \vee 0.\end{aligned}$$

Our first observation is that any fixed point of ϕ is unique and larger than 0, because ϕ is a nonincreasing function and $\phi(0) = q - p_{2|2}\lambda > 0$. Thus solving $\phi(r) = r$ is equivalent to solving

$$q - r + \bar{p}(r \wedge p_{1|1,2}\lambda) - p_{2|2}\lambda = r.$$

If r satisfies this equation, then

$$\begin{aligned}r &= \frac{q - p_{2|2}\lambda + \bar{p}(r \wedge p_{1|1,2}\lambda)}{2} \\ &\leq \frac{2p_{1|1,2}\lambda + p_{2|1,2}\lambda - p_{2|2}\lambda + \bar{p}(r \wedge p_{1|1,2}\lambda)}{2} \\ &= p_{1|1,2}\lambda + \frac{\bar{p}((r \wedge p_{1|1,2}\lambda) - p_{1|1,2}\lambda)}{2} \\ &\leq p_{1|1,2}\lambda.\end{aligned}$$

The first inequality above follows from the conditions on q in the moderate-inventory regime, while the second inequality exploits that $\bar{p} \in (0, 1)$. From these considerations it follows that $r^* \leq p_{1|1,2}\lambda$, so we can determine the fixed point r^* by solving

$$q - r + \bar{p}r - p_{2|2}\lambda = r.$$

We conclude that r^* is given by (EC.5).

We proceed to show that r^* is a globally attracting fixed point for ϕ . The map ϕ is piecewise affine and changes slope once or twice, depending on the parameter choices. If $q \leq (p_{1|1,2} + p_{2|1,2})\lambda$, then $r^\dagger := (q - p_{2|2}\lambda)/(1 - \bar{p}) \leq p_{1|1,2}\lambda$ is the smallest value for which ϕ equals 0. That means that ϕ changes slope once at the point $r^\dagger$ (from slope $-1 + \bar{p}$ to slope 0). For later use we also define $r^\circ := r^\dagger$ in this case. If $q > (p_{1|1,2} + p_{2|1,2})\lambda$, then $r^\dagger := q - p_{2|1,2}\lambda > p_{1|1,2}\lambda$ is the smallest value for which ϕ equals 0. That means that ϕ changes slope twice: at the point $r^\circ := p_{1|1,2}\lambda$ (from slope $-1 + \bar{p}$ to slope -1) and at the point $r^\dagger$ (from slope -1 to slope 0).

For all parameter choices it holds that $0 < r^* \leq r^\circ \leq r^\dagger \leq q$. Now assume that $y_1 < r^* < y_2$. If $y_2 \leq r^\circ$, then

$$\frac{\phi(y_2) - \phi(y_1)}{y_2 - y_1} = \frac{(-1 + \bar{p})(y_2 - y_1)}{y_2 - y_1} = -1 + \bar{p}.$$

If $r^\circ \leq y_2 \leq r^\dagger$, then

$$\begin{aligned} \frac{\phi(y_2) - \phi(y_1)}{y_2 - y_1} &= \frac{\phi(y_2) - \phi(r^\circ)}{y_2 - y_1} + \frac{\phi(r^\circ) - \phi(y_1)}{y_2 - y_1} \\ &= \frac{y_2 - r^\circ}{y_2 - y_1} + \frac{(-1 + \bar{p})(r^\circ - y_1)}{y_2 - y_1} \\ &= 1 + \frac{y_1 - r^\circ}{y_2 - y_1} + \frac{(-1 + \bar{p})(r^\circ - y_1)}{y_2 - y_1}. \end{aligned}$$

Because $y_1 - r^\circ < 0$ and $-1 + \bar{p} < 0$, the latter expression is maximized for $y_2 = r^\circ$. This implies that

$$\frac{\phi(y_2) - \phi(y_1)}{y_2 - y_1} \leq 1 + \frac{y_1 - r^\circ}{r^\circ - y_1} + \frac{(-1 + \bar{p})(r^\circ - y_1)}{r^\circ - y_1} = -1 + \bar{p}.$$

If $r^\dagger \leq y_2$, then

$$\frac{\phi(y_2) - \phi(y_1)}{y_2 - y_1} = \frac{\phi(r^\dagger) - \phi(y_1)}{y_2 - y_1} \leq \frac{\phi(r^\dagger) - \phi(y_1)}{r^\dagger - y_1} \leq -1 + \bar{p}.$$

Based on these inequalities, we conclude that

$$\frac{|\phi(y_2) - \phi(y_1)|}{|y_2 - y_1|} \leq 1 - \bar{p} < 1 \quad \text{if} \quad y_1 < r^* < y_2. \quad (\text{EC.7})$$

Now suppose we have a sequence $y_0, y_1, y_2, \dots$ with $y_{i+1} = \phi(y_i)$ for $i \geq 0$. We first recall the inequalities $\phi(r) \geq r^*$ for $r \leq r^*$ and $\phi(r) \leq r^*$ for $r \geq r^*$, which follow from ϕ being nonincreasing. Thus $y_i \leq r^*$ implies $y_{i+1} \geq r^*$ and $y_i \geq r^*$ implies $y_{i+1} \leq r^*$. Combining this with (EC.7), we see that

$$|y_{i+1} - y_i| = |\phi(y_i) - \phi(y_{i-1})| \leq (1 - \bar{p})|y_i - y_{i-1}|.$$

Because $(1 - \bar{p}) < 1$, the sequence $y_0, y_1, y_2, \dots$ is a Cauchy sequence in $[0, q]$. With r^* being the only fixed point of ϕ , we conclude that r^* is the limit of this sequence.

EC.2.5. Proof Lemma EC.4

Lemma EC.3 demonstrates that, under the assumptions of the moderate-inventory regime $\lambda p_{2|2} < q < 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$, the fixed point r^* of ϕ is globally attracting. Consequently, the limiting dynamical system of $(\frac{1}{m}\mathbf{N}^{(m)}(t) : t \in \mathbb{N}_0)$ has a globally attracting fixed point $(r^* - s_1(r^*), r^*)$. The results of Benaim and Le Boudec (2011) then imply that the scaled inventory vector in stationarity $\frac{1}{m}\mathbf{N}^{(m)}$ converges weakly to $(r^* - s_1(r^*), r^*)$ in the moderate inventory regime. Since the state space of $\frac{1}{m}\mathbf{N}^{(m)}$ is bounded, the expectation of $\frac{1}{m}\mathbf{N}^{(m)}$ converges to $(r^* - s_1(r^*), r^*) = (\bar{r} - s_1(\bar{r}), \bar{r})$ as well.

It remains to show that the latter type of convergence also holds in the scarce-inventory regime and the abundant-inventory regime. As a first step, consider the following dynamics.

prod.	begin period 1	end period 1	begin period 2	end period 2
1	$R^{(m)}$	$R^{(m)} - S_1(1)$	$q^{(m)} - R^{(m)} - S_2(1)$	$q^{(m)} - R^{(m)} - S_2(1) - S_1(2)$
2	$q^{(m)} - R^{(m)}$	$q^{(m)} - R^{(m)} - S_2(1)$	$R^{(m)} + S_2(1)$	$R^{(m)} + S_2(1) - S_2(2)$

At the beginning of period 1, the system starts in stationarity, so $R^{(m)}$ follows the stationary distribution of product 1 at the beginning of a period. Then $S_1(1)$ and $S_2(1)$ are the random variables denoting the sales of product 1 and 2 during period 1, respectively. The random variables $S_1(2)$ and $S_2(2)$ are defined similarly. Because the system starts in stationarity, we know that

$$\mathbf{N}^{(m)} \stackrel{d}{=} (R^{(m)} - S_1(1), q^{(m)} - R^{(m)} - S_2(1)) \stackrel{d}{=} (q^{(m)} - R^{(m)} - S_2(1) - S_1(2), R^{(m)} + S_2(1) - S_2(2)),$$

$$S_1(1) \stackrel{d}{=} S_1(2),$$

$$S_2(1) \stackrel{d}{=} S_2(2),$$

so

$$\mathbb{E}[\mathbf{N}_1^{(m)}] = \frac{\mathbb{E}[R^{(m)} - S_1(1)] + \mathbb{E}[q^{(m)} - R^{(m)} - S_2(1) - S_1(2)]}{2} = \frac{q^{(m)} - 2\mathbb{E}[S_1(1)] - \mathbb{E}[S_2(1)]}{2}$$

and

$$\mathbb{E}[\mathbf{N}_2^{(m)}] = \frac{\mathbb{E}[q^{(m)} - R^{(m)} - S_2(1)] + \mathbb{E}[R^{(m)} + S_2(1) - S_2(2)]}{2} = \frac{q^{(m)} - \mathbb{E}[S_2(1)]}{2}.$$

Since the performance metrics are directly related to the expectation of $\frac{1}{m}\mathbf{N}^{(m)}$, this shows that convergence of the performance metrics follows from convergence of the scaled sales vector. In both the scarce-inventory regime and the abundant-inventory regime the scaled sales vector converges, due to the following comparison arguments.

As for the scarce-inventory regime $q \leq \lambda p_{2|2}$, the order-up-to level q is smaller than in the moderate-inventory regime. Thus the sales of product 1 in the scarce-inventory regime are (first-order) stochastically

dominated by the sales of the same product in the moderate-inventory regime. Under our scaling, the limiting sales of product 1 approach 0 when we approach the boundary case $q = p_{2|2}\lambda$, so we know that the sales of product 1 also converge to 0 in the scarce-inventory regime. Then the scaled initial inventory vector and the sales vector both converge to $(0, q)$ in equilibrium, so $\frac{1}{m}\mathbf{N}^{(m)}$ converges to $(0, 0) = (\bar{r} - s_1(\bar{r}), \bar{r})$ in the scarce-inventory regime. We conclude that the expectation of $\frac{1}{m}\mathbf{N}^{(m)}$ also converges to $(0, 0) = (\bar{r} - s_1(\bar{r}), \bar{r})$ in the scarce-inventory regime.

For the abundant-inventory regime $q \geq 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$, the order-up-to level q is larger than in the moderate-inventory regime. The sales in the abundant-inventory regime stochastically dominate the sales in the moderate-inventory regime. At the same time, the sales in the abundant-inventory regime are stochastically dominated by the infinite-inventory case. In the latter case, the sales vector in equilibrium converges weakly to $(\lambda p_{1|1,2}, \lambda p_{2|1,2})$ under our scaling. Because this is also the limit in the boundary case $q = 2\lambda p_{1|1,2} + \lambda p_{2|1,2}$, the scaled sales vector in the abundant inventory regime must converge weakly to this limit in equilibrium, too. We conclude that the expectation of $\frac{1}{m}\mathbf{N}^{(m)}$ converges to $((q - 2\lambda p_{1|1,2} - \lambda p_{2|1,2})/2, (q - \lambda p_{2|1,2})/2) = (\bar{r} - s_1(\bar{r}), \bar{r})$ in the abundant-inventory regime.

EC.3. Proof of Theorem 3

In this section we prove Theorem 3. Denote $W(\cdot)$ to be the “Product Log” function, i.e., $W(\gamma) = \gamma e^{-W(\gamma)}, \forall \gamma > 0$. Furthermore, to emphasize the dependence on the prices of the products with different remaining shelf-lives, we will write $p_{1|1,2}(r_1, r_2)$, $p_{2|1,2}(r_1, r_2)$, and so on to reflect this dependence. We also write $\tau(r_1, r_2)$, $\text{SALES}_1(r_1, r_2)$, $\text{SALES}_2(r_1, r_2)$, and so on to highlight that the other quantities are also dependent on the prices.

We first outline the proof of Theorem 3 by breaking it into two lemmas.

LEMMA EC.5. *If the following condition holds*

$$\frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)}{1 + W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)} < \frac{q}{\lambda},$$

then the waste can be explicitly expressed as

$$\text{WASTE}(\tilde{r}, \tilde{r}) = q - \lambda \cdot \frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)}{1 + W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)}. \quad (\text{EC.8})$$

LEMMA EC.6. *The following inequality holds:*

$$\frac{2 + e^{v_2 - v_1 + s/2 - c/2}}{1 + e^{v_2 - v_1 + s/2 - c/2}} \cdot \frac{W(e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1})}{1 + W(e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1})} > \frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)}{1 + W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)}.$$

We prove these two lemmas as well as Theorem 3 one by one in the subsequent sections.

EC.3.1. Proof of Lemma EC.5

Proof of Lemma EC.5. When we do not give discounts, both products 1 and 2 are offered at the same price, $r_1 = r_2 = r$. First, when we do not have the flexibility to give discounts, the optimal solution $(\tilde{r}, \tilde{r})$ must be in the case such that $\tau(r, r) > 0$. To see this, we discuss $\tau(r, r) = 0$.

If $\tau(r, r) = 0$, then $q \leq \lambda p_{2|2}(r)$ and the objective of the profit maximization problem reduces to

$$\max_{r \in \mathbb{R}^+} r \cdot q,$$

where q is a constant. The objective function of the profit maximization problem is linearly increasing in r .

But as r increases,

$$\lambda p_{2|2}(r) = \lambda \frac{e^{v_2 - r}}{1 + e^{v_2 - r}}$$

decreases. Till $\lambda p_{2|2}(r)$ decreases to smaller than q , this violates $\tau(r, r) = 0$. As a result, as long as r makes $\tau(r, r) = 0$, this cannot be optimal.

Now that we know the optimal solution must be in the case such that $\tau(r, r) > 0$, there are two possible cases: $0 < \tau(r, r) < 1$, and $\tau(r, r) = 1$.

Case 1: Suppose $0 < \tau(r, r) < 1$. Then we have both $q > \lambda p_{2|2}(r)$ and $q < 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$.

Note that

$$\tau(r, r) = \frac{q - \lambda p_{2|2}(r)}{2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)},$$

which implies

$$\begin{aligned} & \tau(r, r)\lambda p_{2|1,2}(r, r) + (1 - \tau(r, r))\lambda p_{2|2}(r) \\ &= \tau(r, r)[\lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)] + \lambda p_{2|2}(r) \\ &= \tau(r, r)[2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)] - 2\tau(r, r)\lambda p_{1|1,2}(r, r) + \lambda p_{2|2}(r) \\ &= q - \lambda p_{2|2}(r) - 2\lambda p_{1|1,2}(r, r)\tau(r, r) + \lambda p_{2|2}(r) \\ &= q - 2\lambda\tau(r, r)p_{1|1,2}(r, r). \end{aligned}$$

With all above, the objective of the profit maximization problem reduces to

$$\begin{aligned}
& \max_{r \in \mathbb{R}^+} (r - c) \tau(r, r) \lambda p_{1|1,2}(r, r) + (r - c) [\tau(r, r) \lambda p_{2|1,2}(r, r) + (1 - \tau(r, r)) \lambda p_{2|2}(r)] \\
&= \max_{r \in \mathbb{R}^+} (r - c) \tau(r, r) \lambda p_{1|1,2}(r, r) + (r - c) q - (r - c) 2\lambda \tau(r, r) p_{1|1,2}(r, r) \\
&= \max_{r \in \mathbb{R}^+} (r - c) \cdot [q - \tau(r, r) \lambda p_{1|1,2}(r, r)]
\end{aligned}$$

where c, q, λ are all constants. Plugging in the expressions of the purchase probabilities, the objective of the profit maximization problem can be further expressed as

$$\begin{aligned}
& \max_{r \in \mathbb{R}^+} (r - c) \cdot [q - \tau(r, r) \lambda p_{1|1,2}(r, r)] \\
&= \max_{r \in \mathbb{R}^+} (r - c) \cdot \left[q - \frac{q - \lambda p_{2|2}(r)}{2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)} \cdot \lambda p_{1|1,2}(r, r) \right] \\
&= \max_{r \in \mathbb{R}^+} (r - c) \cdot \left[q - \left(q - \lambda \frac{e^{v_2 - r}}{1 + e^{v_2 - r}} \right) \frac{1 + e^{v_2 - r}}{2 + e^{v_2 - r}} \right] \\
&= \max_{r \in \mathbb{R}^+} (r - c) \cdot \left[q \cdot \frac{1}{2 + e^{v_2 - r}} + \lambda \cdot \frac{e^{v_2 - r}}{2 + e^{v_2 - r}} \right].
\end{aligned}$$

So in Case 1, we solve the following optimization problem

$$\begin{aligned}
& \max_{r \in \mathbb{R}^+} (r - c) \cdot \left[q \cdot \frac{1}{2 + e^{v_2 - r}} + \lambda \cdot \frac{e^{v_2 - r}}{2 + e^{v_2 - r}} \right] \\
& \text{s.t. } 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) > q \\
& \lambda p_{2|2}(r, r) < q.
\end{aligned}$$

Now define $f_1(r) := (r - c) \cdot \left[q \cdot \frac{1}{2 + e^{v_2 - r}} + \lambda \cdot \frac{e^{v_2 - r}}{2 + e^{v_2 - r}} \right]$, and observe that

$$f_1'(r) = \frac{q - 2\lambda}{(2 + e^{v_2 - r})^2} \cdot (2 + e^{v_2 - r} + (r - c)e^{v_2 - r}) + \lambda.$$

Note that, if $q \geq 2\lambda$, then $f_1'(r) \geq \lambda > 0$, implying f_1 is an increasing function. If $q \geq 2\lambda$ then the optimal solution cannot be in Case 1. On the other hand, if $q < 2\lambda$, then $f_1'(r)$ may or may not be equal to zero. For example, in Figure EC.1, there does not exist an optimal solution in Case 1. On the other hand, in Figure EC.2, there might exist an optimal solution in Case 1.

Case 2: Suppose $\tau(r, r) = 1$. Then we have both $q > \lambda p_{2|2}(r)$ and $q \geq 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$.

The objective of the profit maximization problem reduces to

$$\max_{r \in \mathbb{R}^+} (r - s) \lambda p_{1|1,2}(r, r) + \left(r - \frac{s}{2} - \frac{c}{2} \right) \lambda p_{2|1,2}(r, r) + \frac{s - c}{2} q,$$

where c, q, λ are all constants.

Note that $2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) \geq \lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) \geq \lambda p_{2|2}(r)$, and $\lambda > 0$. So it suffices to ensure that $q \geq 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$. Plugging in the expressions of the purchase probabilities, the profit maximization problem that we solve under Case 2 reduces to

$$\begin{aligned} \max_{r \in \mathbb{R}^+} & \frac{\lambda(r-s)e^{v_1-r}}{1+e^{v_1-r}+e^{v_2-r}} + \frac{\lambda(r-s/2-c/2)e^{v_2-r}}{1+e^{v_1-r}+e^{v_2-r}} + \frac{s-c}{2} \cdot q \\ \text{s.t. } & 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) \leq q. \end{aligned}$$

Now we denote $f_2(r) := \frac{\lambda(r-s)e^{v_1-r}}{1+e^{v_1-r}+e^{v_2-r}} + \frac{\lambda(r-s/2-c/2)e^{v_2-r}}{1+e^{v_1-r}+e^{v_2-r}} + \frac{s-c}{2} \cdot q$, then

$$\begin{aligned} f_2'(r) &= \lambda \cdot \frac{e^{v_1-r}(1-r+s)(1+e^{v_1-r}+e^{v_2-r}) + (r-s)e^{v_1-r}(e^{v_1-r}+e^{v_2-r})}{(1+e^{v_1-r}+e^{v_2-r})^2} \\ &\quad + \frac{e^{v_2-r}(1-r+s/2+c/2)(1+e^{v_1-r}+e^{v_2-r}) + (r-s/2-c/2)e^{v_2-r}(e^{v_1-r}+e^{v_2-r})}{(1+e^{v_1-r}+e^{v_2-r})^2} \\ &= \lambda \cdot \frac{(e^{v_1-r}+e^{v_2-r})(1+e^{v_1-r}+e^{v_2-r}) - (r-s)e^{v_1-r} - (r-s/2-c/2)e^{v_2-r}}{(1+e^{v_1-r}+e^{v_2-r})^2} \\ &= \lambda \cdot \frac{e^{-r}[(e^{v_1}+e^{v_2})^2e^{-r} - r(e^{v_1}+e^{v_2}) + (1+s)e^{v_1} + (1+s/2+c/2)e^{v_2}]}{(1+e^{v_1-r}+e^{v_2-r})^2}. \end{aligned}$$

Let $r = \tilde{r}_2$ be one solution that satisfies the above first order condition. Note that the subscript 2 in $\tilde{r}_2$ is aligned with the subscript 2 in $f_2(r)$. Setting $f_2'(\tilde{r}_2) = 0$ and rearranging terms, we have

$$(e^{v_1}+e^{v_2})e^{-\tilde{r}_2} - \tilde{r}_2 + \left(1 + \frac{s}{2} + \frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}\right) = 0.$$

So the first order condition then leads to

$$\begin{aligned} \tilde{r}_2 - \left(1 + \frac{s}{2} + \frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}\right) &= e^{-\tilde{r}_2}(e^{v_1}+e^{v_2}) \\ &= e^{-\tilde{r}_2+1+\frac{s}{2}+\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \right) \end{aligned}$$

By the definition of Product Log function,

$$\tilde{r}_2 = W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \right) + 1 + \frac{s}{2} + \frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}. \quad (\text{EC.9})$$

Note that, in the square bracket on the numerator of $f_2'(r)$, $(e^{v_1}+e^{v_2})^2e^{-r} - r(e^{v_1}+e^{v_2})$ is monotone decreasing in r . So when $r < \tilde{r}_2$, $f_2'(r) > 0$; and when $r > \tilde{r}_2$, $f_2'(r) < 0$. Thus, $f_2(r)$ is a uni-modal function with one unique maximum at $r = \tilde{r}_2$. See the red curves in Figures EC.1 and EC.2.

Now let us check if the constraint is satisfied in order for $\tau(\tilde{r}_2, \tilde{r}_2)$ to be equal to 1. Note that

$$\begin{aligned} 2p_{1|1,2}(\tilde{r}_2, \tilde{r}_2) + p_{2|1,2}(\tilde{r}_2, \tilde{r}_2) &= \frac{2e^{v_1-\tilde{r}_2} + e^{v_2-\tilde{r}_2}}{1+e^{v_1-\tilde{r}_2}+e^{v_2-\tilde{r}_2}} \\ &= \frac{2+e^{v_2-v_1}}{1+e^{v_2-v_1}} \cdot \frac{W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \right)}{1+W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \right)}. \end{aligned}$$

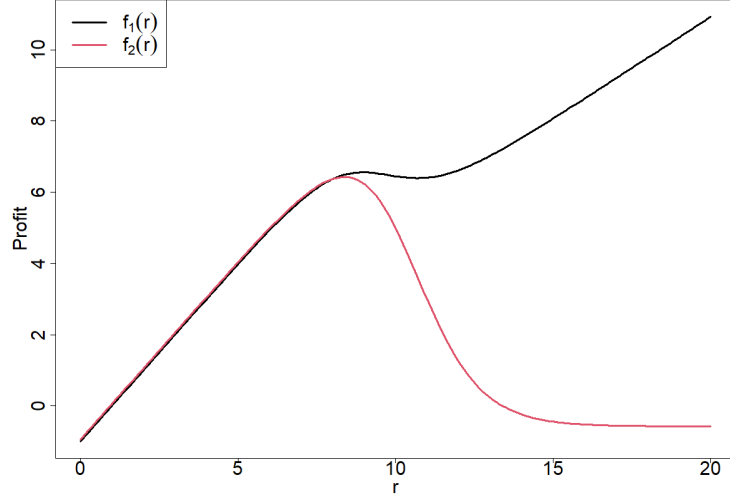


Figure EC.1 Illustration of the profit function as a function of price. In this figure, the black curve stands for $f_1(\cdot)$ and the red curve stands for $f_2(\cdot)$. The parameters are $c = 1, s = 0, v_1 = 9, v_2 = 10, q = 1.15, \lambda = 1$.

So in Case 2, if

$$\frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}{1 + W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)} \leq q/\lambda,$$

then (EC.9) is the optimal solution.

Furthermore, if (EC.9) is the optimal solution, the waste can be expressed as

$$\begin{aligned} \text{WASTE}(\tilde{r}_2, \tilde{r}_2) &= \frac{1}{2} (q - 2\lambda p_{1|1,2}(\tilde{r}_2, \tilde{r}_2) - \lambda p_{2|1,2}(\tilde{r}_2, \tilde{r}_2))^+ \\ &= q - \lambda \cdot \frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}{1 + W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}. \end{aligned}$$

Discussion of the optimal price. We argue that if

$$\frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}{1 + W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)} \leq q/\lambda$$

holds, then the optimal price must be under Case 2.

So far we have defined $f_1(\cdot)$ and $f_2(\cdot)$. Now let us define

$$f_0(r) = \frac{2e^{v_1 - r} + e^{v_2 - r}}{1 + e^{v_1 - r} + e^{v_2 - r}}.$$

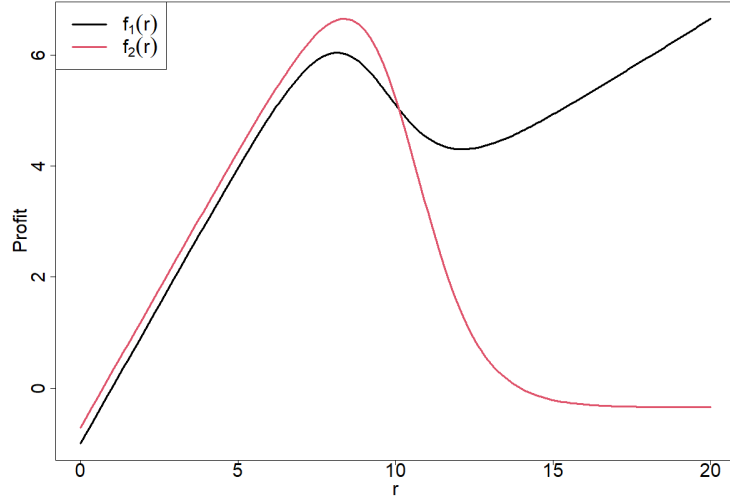


Figure EC.2 Illustration of the profit function as a function of price. In this figure, the black curve stands for $f_1(\cdot)$ and the red curve stands for $f_2(\cdot)$. The parameters are $c = 1, s = 0, v_1 = 9, v_2 = 10, q = 0.7, \lambda = 1$.

Note that $\forall r \in \mathbb{R}, 0 < f_0(r) < 2$, and that $f_0(r)$ is a monotone decreasing function. Also note that $q/\lambda \in (0, 2)$, because otherwise the optimal solution must be in Case 2. Denote $\tilde{r}_0$ to be the unique solution to $f_0(\tilde{r}_0) = q/\lambda$. Note that $\tilde{r}_0$ could be negative.

First, observe that $f_1(\tilde{r}_0) = f_2(\tilde{r}_0)$. To see this, note that $\frac{2e^{v_1-\tilde{r}_0}+e^{v_2-\tilde{r}_0}}{1+e^{v_1-\tilde{r}_0}+e^{v_2-\tilde{r}_0}} = \frac{q}{\lambda}$, and so $\frac{q-2\lambda}{\lambda} = \frac{-2-e^{v_2-\tilde{r}_0}}{1+e^{v_1-\tilde{r}_0}+e^{v_2-\tilde{r}_0}}$. Then using this we have

$$f_1(\tilde{r}_0) = (\tilde{r}_0 - c) \cdot \left[\lambda + \frac{q - 2\lambda}{2 + e^{v_2 - \tilde{r}_0}} \right] = (\tilde{r}_0 - c) \cdot \frac{\lambda(e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0})}{1 + e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0}}.$$

Moreover, we have

$$f_2(\tilde{r}_0) = \frac{\lambda(\tilde{r}_0 - s)e^{v_1 - \tilde{r}_0}}{1 + e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0}} + \frac{\lambda(\tilde{r}_0 - s/2 - c/2)e^{v_2 - \tilde{r}_0}}{1 + e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0}} + \frac{s - c}{2} \cdot q = \frac{\lambda(\tilde{r}_0 - c)(e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0})}{1 + e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0}}.$$

Combining above we see that $f_1(\tilde{r}_0) = f_2(\tilde{r}_0) = \frac{\lambda(\tilde{r}_0 - c)(e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0})}{1 + e^{v_1 - \tilde{r}_0} + e^{v_2 - \tilde{r}_0}}$. This corresponds to the crossing point of the black and red curves in both Figures EC.1 and EC.2. Note that, this observation does not exclude the possibility that the black and red curves have more than one crossing point.

Second, define the following function:

$$f(r) = \begin{cases} f_1(r) = (r - c) \cdot \left[\lambda + \frac{q - 2\lambda}{2 + e^{v_2 - r}} \right], & r \leq \tilde{r}_0 \\ f_2(r) = \frac{\lambda(r - s)e^{v_1 - r}}{1 + e^{v_1 - r} + e^{v_2 - r}} + \frac{\lambda(r - s/2 - c/2)e^{v_2 - r}}{1 + e^{v_1 - r} + e^{v_2 - r}} + \frac{s - c}{2} \cdot q, & r > \tilde{r}_0 \end{cases}$$

We know that the maximizer of this function $\arg \max_{r \in \mathbb{R}^+} f(r)$ is the profit-maximizing price.

Third, we wish to argue that when $r \leq \tilde{r}_0$, $f_2(r) \geq f_1(r)$. To see this, note that $f_0(r)$ is a monotone decreasing function. So for any $r \leq \tilde{r}_0$,

$$f_0(r) = \frac{2e^{v_1-r} + e^{v_2-r}}{1 + e^{v_1-r} + e^{v_2-r}} \geq f_0(\tilde{r}_0) = \frac{q}{\lambda}. \quad (\text{EC.10})$$

Then we have

$$\begin{aligned} f_2(r) &= \frac{\lambda(r-s)e^{v_1-r}}{1 + e^{v_1-r} + e^{v_2-r}} + \frac{\lambda(r-s/2-c/2)e^{v_1-r}}{1 + e^{v_1-r} + e^{v_2-r}} + \frac{s-c}{2}q \\ &\geq \frac{\lambda(r-s)e^{v_1-r}}{1 + e^{v_1-r} + e^{v_2-r}} + \frac{\lambda(r-s/2-c/2)e^{v_1-r}}{1 + e^{v_1-r} + e^{v_2-r}} + \frac{s-c}{2} \cdot \lambda \cdot \frac{2e^{v_1-r} + e^{v_2-r}}{1 + e^{v_1-r} + e^{v_2-r}} \\ &= \lambda(r-c) \frac{e^{v_1-r} + e^{v_2-r}}{1 + e^{v_1-r} + e^{v_2-r}} \\ &= \lambda(r-c) \frac{\frac{2e^{v_1-r} + e^{v_2-r}}{1 + e^{v_1-r} + e^{v_2-r}} + e^{v_2-r}}{2 + e^{v_2-r}} \\ &\geq \lambda(r-c) \frac{\frac{q}{\lambda} + e^{v_2-r}}{2 + e^{v_2-r}} \\ &= f_1(r), \end{aligned}$$

where the second inequality is using (EC.10) and because $s - c < 0$; the second and third equalities are re-arranging terms; the last inequality is again using (EC.10).

Fourth and finally, we argue that if

$$\frac{2 + e^{v_2-v_1}}{1 + e^{v_2-v_1}} \cdot \frac{W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} \right)}{1 + W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} \right)} < \frac{q}{\lambda},$$

then $\tilde{r}_2$ from (EC.9) is the optimal solution to the profit maximization problem. To see this, note that

$$f_0(\tilde{r}_0) = \frac{q}{\lambda} > \frac{2 + e^{v_2-v_1}}{1 + e^{v_2-v_1}} \cdot \frac{W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} \right)}{1 + W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} \right)} = f_0(\tilde{r}_2).$$

Because $f_0(\cdot)$ is a decreasing function, $\tilde{r}_0 < \tilde{r}_2$. Next, note that $f_2(\cdot)$ is a uni-modal function with a unique global maximum at $\tilde{r}_2$. So for any $r \leq \tilde{r}_0 < \tilde{r}_2$,

$$f(r) = f_1(r) \leq f_2(r) < f_2(\tilde{r}_2).$$

This implies, under the condition that

$$\frac{2 + e^{v_2-v_1}}{1 + e^{v_2-v_1}} \cdot \frac{W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} \right)}{1 + W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} \right)} < \frac{q}{\lambda},$$

$r = \tilde{r}_2$ is the unique global maximum for function $f(r)$, thus the maximizer of the profit maximization problem.

Finally, if the optimal solution is in Case 2, the waste can be expressed as

$$\begin{aligned} \text{WASTE}(\tilde{r}, \tilde{r}) &= \frac{1}{2} (q - 2\lambda p_{1|1,2}(\tilde{r}, \tilde{r}) - \lambda p_{2|1,2}(\tilde{r}, \tilde{r}))^+ \\ &= q - \lambda \cdot \frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)}{1 + W\left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right)}, \end{aligned}$$

which concludes the proof. $\square$

EC.3.2. Proof of Lemma EC.6

Proof of Lemma EC.6. First, observe that the ratio

$$\frac{2 + e^{v_2 - v_1 - x}}{1 + e^{v_2 - v_1 - x}}$$

is decreasing in x . Then from $s < c$ we have

$$\frac{2 + e^{v_2 - v_1 + s/2 - c/2}}{1 + e^{v_2 - v_1 + s/2 - c/2}} > \frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} > 0.$$

Second, observe that the Product Log function $W(\cdot)$ is an increasing function on $[0, +\infty)$. So $\frac{W(x)}{1+W(x)}$ is also an increasing function on $[0, +\infty)$. Next, to compare $e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1}$ that appears in (EC.15) and $e^{v_1 - 1 - s/2 - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - s/2 - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}$ that appears in (EC.8), define

$$h(v_1, v_2, s, c) = e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1} - e^{v_1 - 1 - s/2 - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} - e^{v_2 - 1 - s/2 - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}.$$

We focus on the dependence of $h(v_1, v_2, s, c)$ on c .

$$\begin{aligned} \frac{\partial h(v_1, v_2, s, c)}{\partial c} &= -\frac{1}{2} e^{v_2 - s/2 - c/2 - 1} + \frac{e^{v_2}}{2(e^{v_1} + e^{v_2})} e^{v_1 - 1 - s/2 - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + \frac{e^{v_2}}{2(e^{v_1} + e^{v_2})} e^{v_2 - 1 - s/2 - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \\ &= -\frac{1}{2} e^{v_2 - s/2 - c/2 - 1} + \frac{1}{2} e^{v_2 - 1 - s/2 - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \\ &= \frac{e^{v_2 - s/2 - c/2 - 1}}{2} \cdot \left(e^{\frac{(c-s)e^{v_2}}{2(e^{v_1} + e^{v_2})}} - 1 \right) \\ &> 0, \end{aligned}$$

where the last inequality is because $c > s$. So $h(v_1, v_2, s, c)$ is increasing in c . Then we have

$$h(v_1, v_2, s, c) > h(v_1, v_2, s, s) = 0.$$

Then using the fact that $\frac{W(x)}{1+W(x)}$ is an increasing function, we know that

$$\frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1+W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})} > \frac{W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}}\right)}{1+W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}}\right)} > 0.$$

Combining the two observations, we have

$$\begin{aligned} \frac{2+e^{v_2-v_1+s/2-c/2}}{1+e^{v_2-v_1+s/2-c/2}} \cdot \frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1+W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})} > \\ \frac{2+e^{v_2-v_1}}{1+e^{v_2-v_1}} \cdot \frac{W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}}\right)}{1+W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}}\right)}. \end{aligned}$$

□

EC.3.3. Proof of Theorem 3

Proof of Theorem 3. We distinguish the following 4 cases:

1. $\text{WASTE}(r_1^*, r_2^*) > 0$, $\text{WASTE}(\tilde{r}, \tilde{r}) > 0$;
2. $\text{WASTE}(r_1^*, r_2^*) > 0$, $\text{WASTE}(\tilde{r}, \tilde{r}) = 0$;
3. $\text{WASTE}(r_1^*, r_2^*) = 0$, $\text{WASTE}(\tilde{r}, \tilde{r}) > 0$;
4. $\text{WASTE}(r_1^*, r_2^*) = 0$, $\text{WASTE}(\tilde{r}, \tilde{r}) = 0$.

For cases 3 and 4, because $\text{WASTE}(r_1^*, r_2^*) = 0$, we already know that $\text{WASTE}(\tilde{r}, \tilde{r}) \geq 0 = \text{WASTE}(r_1^*, r_2^*)$.

For cases 1 and 2, we start with $\text{WASTE}(r_1^*, r_2^*) > 0$. In the case when $\text{WASTE}(r_1^*, r_2^*) > 0$, we first ignore the constraint $r_2 \geq r_1 \geq 0$ and solve the optimal r_1^* and r_2^* . We then check that r_1^* and r_2^* automatically satisfies the constraint $r_2 \geq r_1 \geq 0$.

We solve the (unconstrained) optimal r_1^* and r_2^* below. First, when we choose both r_1 and r_2 with the flexibility to give discounts, the optimal solution (r_1^*, r_2^*) must be in the case such that $\tau(r_1, r_2) = 1$. To see this, we discuss $\tau(r_1, r_2) = 0$ and $0 < \tau(r_1, r_2) < 1$ cases separately.

If $\tau(r_1, r_2) = 0$, we have $q \leq \lambda p_{2|2}(r_2)$. Then the objective function of the profit maximization problem reduces to

$$\max_{r_2 \in \mathbb{R}^+} r_2 \cdot q,$$

where q is a constant. The objective function of the profit maximization problem is linearly increasing in r_2 . But as r_2 increases, the quantity

$$\lambda p_{2|2}(r_2) = \lambda \frac{e^{v_2-r_2}}{1+e^{v_2-r_2}}$$

decreases. Till $\lambda p_{2|2}(r_2)$ decreases to smaller than q , this violates $\tau(r_1, r_2) = 0$. As a result, as long as r_1 and r_2 makes $\tau(r_1, r_2) = 0$, this cannot be optimal.

If $0 < \tau(r_1, r_2) < 1$, we have both $q > \lambda p_{2|2}(r_2)$ and $q < 2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2)$. Note that

$$\tau(r_1, r_2) = \frac{q - \lambda p_{2|2}(r_2)}{2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) - \lambda p_{2|2}(r_2)},$$

which implies

$$\begin{aligned} & \tau(r_1, r_2)\lambda p_{2|1,2}(r_1, r_2) + (1 - \tau(r_1, r_2))\lambda p_{2|2}(r_2) \\ &= \tau(r_1, r_2)(\lambda p_{2|1,2}(r_1, r_2) - \lambda p_{2|2}(r_2)) + \lambda p_{2|2}(r_2) \\ &= \tau(r_1, r_2)(2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) - \lambda p_{2|2}(r_2)) - 2\tau(r_1, r_2)\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|2}(r_2) \\ &= q - \lambda p_{2|2}(r_2) - 2\lambda p_{1|1,2}(r_1, r_2)\tau(r_1, r_2) + \lambda p_{2|2}(r_2) \\ &= q - 2\lambda\tau(r_1, r_2)p_{1|1,2}(r_1, r_2). \end{aligned}$$

With all above, the profit maximization problem reduces to

$$\begin{aligned} & \max_{r_1, r_2 \in \mathbb{R}^+} (r_1 - c)\tau(r_1, r_2)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - c) [\tau(r_1, r_2)\lambda p_{2|1,2}(r_1, r_2) + (1 - \tau(r_1, r_2))\lambda p_{2|2}(r_2)] \\ &= \max_{r_1, r_2 \in \mathbb{R}^+} (r_1 - c)\tau(r_1, r_2)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - c)q - (r_2 - c)2\lambda\tau(r_1, r_2)p_{1|1,2}(r_1, r_2) \\ &= \max_{r_1, r_2 \in \mathbb{R}^+} \tau(r_1, r_2)\lambda p_{1|1,2}(r_1, r_2)(r_1 - 2r_2 + c) + (r_2 - c)q \end{aligned}$$

where c, q, λ are all constants.

Note that, by plugging in the expressions of the choice probabilities,

$$\begin{aligned} & \tau(r_1, r_2)\lambda p_{1|1,2}(r_1, r_2)(r_1 - 2r_2 + c) + (r_2 - c)q \\ &= (q - \lambda p_{2|2}(r_2)) \frac{p_{1|1,2}(r_1, r_2)}{2p_{1|1,2}(r_1, r_2) + p_{2|1,2}(r_1, r_2) - p_{2|2}(r_2)} (r_1 - 2r_2 + c) + (r_2 - c)q \\ &= (q - \lambda p_{2|2}(r_2)) \frac{1 + e^{v_2 - r_2}}{2 + e^{v_2 - r_2}} (r_1 - 2r_2 + c) + (r_2 - c)q. \end{aligned}$$

Because $q > \lambda p_{2|2}(r_2)$, we know that the objective function of the profit maximization problem is monotone increasing in r_1 . But as r_1 increases, the quantity

$$2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) = \lambda \cdot \frac{2e^{v_1 - r_1} + e^{v_2 - r_2}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}}$$

decreases. Till $2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2)$ decreases to q , this violates $\tau(r_1, r_2) < 1$. As a result, as long as r_1 and r_2 makes $\tau(r_1, r_2) < 1$, this cannot be optimal.

From above, we know the optimal solution (r_1^*, r_2^*) must be in the case such that $\tau(r_1, r_2) = 1$. When $\tau(r_1, r_2) = 1$, we have both $q > \lambda p_{2|2}(r_2)$ and $q \geq 2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2)$.

Then the profit maximization problem reduces to

$$\begin{aligned} & \max_{r_1, r_2 \in \mathbb{R}^+} (r_1 - c)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - c)\lambda p_{2|1,2} + \frac{s-c}{2}(q - 2\lambda p_{1|1,2}(r_1, r_2) - \lambda p_{2|1,2}(r_1, r_2)) \\ &= \max_{r_1, r_2 \in \mathbb{R}^+} (r_1 - s)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - \frac{s}{2} - \frac{c}{2})\lambda p_{2|1,2}(r_1, r_2) + \frac{s-c}{2}q, \end{aligned}$$

where c, q, λ are all constants.

Plugging in the expressions of MNL we re-write the profit maximization problem as:

$$\begin{aligned} (r_1^*, r_2^*) &= \arg \max_{r_1, r_2 \in \mathbb{R}^+} \frac{\lambda(r_1 - s)e^{v_1 - r_1}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}} + \frac{\lambda(r_2 - \frac{s}{2} - \frac{c}{2})e^{v_2 - r_2}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}} + \frac{s-c}{2}q \\ &\text{s.t. } 2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) \leq q. \end{aligned}$$

First, note that the optimal solution must be non-negative, even without the constraint $r_1, r_2 \in \mathbb{R}^+$. Second, note that when the optimal solution to the above profit maximization problem is on the boundary, then $2\lambda p_{1|1,2}(r_1^*, r_2^*) + \lambda p_{2|1,2}(r_1^*, r_2^*) = q$, which suggests that $\text{WASTE}(r_1^*, r_2^*) = 0$. But $\text{WASTE}(r_1^*, r_2^*) > 0$ so the optimal solution is not on the boundary. Then due to the KKT condition, we check the necessary condition below.

Now denote

$$g(r_1, r_2) = \frac{(r_1 - s)e^{v_1 - r_1}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}} + \frac{(r_2 - \frac{s}{2} - \frac{c}{2})e^{v_2 - r_2}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}},$$

and we can check the first order derivatives. First for r_1 ,

$$\begin{aligned} & \frac{\partial g(r_1, r_2)}{\partial r_1} \\ &= \frac{[e^{v_1 - r_1}(1 - r_1)](1 + e^{v_1 - r_1} + e^{v_2 - r_2}) + (r_1 - s)e^{v_1 - r_1}e^{v_1 - r_1}}{(1 + e^{v_1 - r_1} + e^{v_2 - r_2})^2} + \frac{(r_2 - s/2 - c/2)e^{v_2 - r_2}e^{v_1 - r_1}}{(1 + e^{v_1 - r_1} + e^{v_2 - r_2})^2} \\ &= \frac{e^{v_1 - r_1}}{(1 + e^{v_1 - r_1} + e^{v_2 - r_2})^2} [1 - r_1 + s + e^{v_2 - r_2}(1 - r_1 + r_2 + s/2 - c/2) + e^{v_1 - r_1}]. \end{aligned}$$

Then for r_2 ,

$$\begin{aligned} & \frac{\partial g(r_1, r_2)}{\partial r_2} \\ &= \frac{(r_1 - s)e^{v_1 - r_1}e^{v_2 - r_2}}{(1 + e^{v_1 - r_1} + e^{v_2 - r_2})^2} + \frac{[e^{v_2 - r_2}(1 - r_2 + s/2 + c/2)](1 + e^{v_1 - r_1} + e^{v_2 - r_2}) + (r_2 - s/2 - c/2)e^{v_2 - r_2}e^{v_2 - r_2}}{(1 + e^{v_1 - r_1} + e^{v_2 - r_2})^2} \\ &= \frac{e^{v_2 - r_2}}{(1 + e^{v_1 - r_1} + e^{v_2 - r_2})^2} [1 - r_2 + s/2 + c/2 + e^{v_1 - r_1}(1 + r_1 - r_2 - s/2 + c/2) + e^{v_2 - r_2}]. \end{aligned}$$

Setting $\frac{\partial g(r_1, r_2)}{\partial r_1} \Big|_{\substack{r_1=r_1^* \\ r_2=r_2^*}} = 0$ implies

$$1 + e^{v_1 - r_1^*} + e^{v_2 - r_2^*} = r_1^* - s + e^{v_2 - r_2^*}(r_1^* - r_2^* - s/2 + c/2). \quad (\text{EC.11})$$

And setting $\left. \frac{\partial g(r_1, r_2)}{\partial r_2} \right|_{\substack{r_1=r_1^* \\ r_2=r_2^*}} = 0$ implies

$$1 + e^{v_1 - r_1^*} + e^{v_2 - r_2^*} = r_2^* - s/2 - c/2 - e^{v_1 - r_1^*}(r_1^* - r_2^* - s/2 + c/2). \quad (\text{EC.12})$$

Combining (EC.11) and (EC.12) we have

$$r_1^* - s + e^{v_2 - r_2^*}(r_1^* - r_2^* - s/2 + c/2) = r_2^* - s/2 - c/2 - e^{v_1 - r_1^*}(r_1^* - r_2^* - s/2 + c/2).$$

Rearranging terms leads to

$$(1 + e^{v_1 - r_1^*} + e^{v_2 - r_2^*})(r_1^* - r_2^* - s/2 + c/2) = 0.$$

Because $1 + e^{v_1 - r_1^*} + e^{v_2 - r_2^*} > 0$ we have that $r_1^* = r_2^* + s/2 - c/2$.

Furthermore, if we plug $r_1^* = r_2^* + s/2 - c/2$ into (EC.11) we have

$$1 + e^{v_1 - r_1^*} + e^{v_2 - r_2^*} = r_1^* - s. \quad (\text{EC.13})$$

Re-arranging terms we have

$$\begin{aligned} r_1^* &= 1 + s + e^{v_1 - r_1^*} + e^{v_2 - r_2^*} \\ &= 1 + s + e^{v_1} \cdot e^{-r_1^*} + e^{v_2 + s/2 - c/2} \cdot e^{-r_1^*} \\ &= 1 + s + (e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1}) \cdot e^{-r_1^* + s + 1}, \end{aligned}$$

which implies $(e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1}) \cdot e^{-(r_1^* - s - 1)} = r_1^* - s - 1$. So the optimal solution can be expressed as

$$\begin{aligned} r_1^* &= W(e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1}) + 1 + s, \\ r_2^* &= W(e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1}) + 1 + s/2 + c/2, \end{aligned} \quad (\text{EC.14})$$

where $W(\cdot)$ is the ‘‘Product Log’’ function, i.e., $W(\gamma) = \gamma e^{-W(\gamma)}, \forall \gamma > 0$.

Due to Proposition 2 from Aydin and Porteus (2008) the first order condition is sufficient to showing that (EC.14) is a global maximum.

Now we see that r_1^* and r_2^* automatically satisfy the constraint $r_2 \geq r_1 \geq 0$. So the optimal solutions of the constrained profit maximization problem is the same as the optimal solutions of the unconstrained profit maximization problem, that is, r_1^* and r_2^* will remain the same.

Now we plug in the expression of (EC.14) and use $r_1^* = r_2^* + s/2 - c/2$ to express the waste function. Because $\text{WASTE}(r_1^*, r_2^*) > 0$ is equivalent to $2\lambda p_{1|1,2}(r_1^*, r_2^*) + \lambda p_{2|1,2}(r_1^*, r_2^*) < q$, it follows that

$$\begin{aligned}
 2p_{1|1,2}(r_1^*, r_2^*) + p_{2|1,2}(r_1^*, r_2^*) &= \frac{2e^{v_1-r_1^*} + e^{v_2-r_2^*}}{1 + e^{v_1-r_1^*} + e^{v_2-r_2^*}} \\
 &= \frac{e^{v_1-r_1^*} + r_1^* - 1 - s}{r_1^* - s} \\
 &= \frac{e^{v_1-W(e^{v_1-s-1}+e^{v_2-s/2-c/2-1})-1-s} + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1}) + 1} \\
 &= \frac{e^{v_1-1-s} \frac{W(e^{v_1-s-1}+e^{v_2-s/2-c/2-1})}{e^{v_1-s-1}+e^{v_2-s/2-c/2-1}} + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1}) + 1} \\
 &= \frac{2 + e^{v_2-v_1+s/2-c/2}}{1 + e^{v_2-v_1+s/2-c/2}} \cdot \frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1 + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}.
 \end{aligned}$$

We explain the equalities. The second equality is due to (EC.13). The third equality is due to (EC.14). The fourth equality is due to the definition of the product log function.

So if $\text{WASTE}(r_1^*, r_2^*) > 0$, then using (EC.14) the waste can be explicitly expressed as

$$\text{WASTE}(r_1^*, r_2^*) = q - \lambda \cdot \frac{2 + e^{v_2-v_1+s/2-c/2}}{1 + e^{v_2-v_1+s/2-c/2}} \cdot \frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1 + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})} > 0. \quad (\text{EC.15})$$

On the other hand, if $\text{WASTE}(r_1^*, r_2^*) > 0$, then

$$\frac{2 + e^{v_2-v_1+s/2-c/2}}{1 + e^{v_2-v_1+s/2-c/2}} \cdot \frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1 + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})} < \frac{q}{\lambda}. \quad (\text{EC.16})$$

Combining (EC.16) and Lemma EC.6 we have

$$\frac{2 + e^{v_2-v_1}}{1 + e^{v_2-v_1}} \cdot \frac{W\left(e^{\frac{v_1-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{\frac{v_2-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}\right)}{1 + W\left(e^{\frac{v_1-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{\frac{v_2-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}\right)} < \frac{q}{\lambda}. \quad (\text{EC.17})$$

Then due to Lemma EC.5 the waste can be explicitly expressed as

$$\text{WASTE}(\tilde{r}, \tilde{r}) = q - \lambda \cdot \frac{2 + e^{v_2-v_1}}{1 + e^{v_2-v_1}} \cdot \frac{W\left(e^{\frac{v_1-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{\frac{v_2-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}\right)}{1 + W\left(e^{\frac{v_1-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{\frac{v_2-1-s/2-se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}\right)} > 0,$$

where the inequality is due to (EC.17).

Combining the two expressions of the waste, $\text{WASTE}(r_1^*, r_2^*)$ and $\text{WASTE}(\tilde{r}, \tilde{r})$, we know that whenever $\text{WASTE}(r_1^*, r_2^*) > 0$ is positive, $\text{WASTE}(\tilde{r}, \tilde{r}) > 0$ must be positive as well. So Case 2 does not exist. Further due to Lemma EC.6 we have

$$\text{WASTE}(r_1^*, r_2^*) < \text{WASTE}(\tilde{r}, \tilde{r}),$$

which finishes the proof. $\square$

EC.4. Proof of Proposition 4

Proof of Proposition 4. To prove the first part, note that, when $\text{WASTE}(\tilde{r}, \tilde{r}) > \text{WASTE}(r_1^*, r_2^*) > 0$, the optimal markdown prices can be expressed by (EC.14) and the optimal fixed price can be expressed by (EC.9). Here we re-write them again.

$$\begin{aligned} r_1^* &= W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1}) + 1 + s, \\ r_2^* &= W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1}) + 1 + s/2 + c/2, \\ \tilde{r} &= W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}\right) + 1 + \frac{s}{2} + \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}. \end{aligned}$$

First, we show $r_2^* > \tilde{r}$. Observe that the Product Log function $W(\cdot)$ is an increasing function on $[0, +\infty)$. To compare $e^{v_1-s-1} + e^{v_2-s/2-c/2-1}$ that appears in the expression in r_2^* and $e^{v_1-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}$ that appears in the expression in $\tilde{r}$, define

$$h(v_1, v_2, s, c) = e^{v_1-s-1} + e^{v_2-s/2-c/2-1} - e^{v_1-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} - e^{v_2-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}.$$

We focus on the dependence of $h(v_1, v_2, s, c)$ on c .

$$\begin{aligned} \frac{\partial h(v_1, v_2, s, c)}{\partial c} &= -\frac{1}{2}e^{v_2-s/2-c/2-1} + \frac{e^{v_2}}{2(e^{v_1} + e^{v_2})}e^{v_1-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + \frac{e^{v_2}}{2(e^{v_1} + e^{v_2})}e^{v_2-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \\ &= -\frac{1}{2}e^{v_2-s/2-c/2-1} + \frac{1}{2}e^{v_2-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \\ &= \frac{e^{v_2-s/2-c/2-1}}{2} \cdot \left(e^{\frac{(c-s)e^{v_2}}{2(e^{v_1}+e^{v_2})}}} - 1 \right) \\ &> 0, \end{aligned}$$

where the last inequality is because $c > s$. So $h(v_1, v_2, s, c)$ is increasing in c . Then we have

$$h(v_1, v_2, s, c) > h(v_1, v_2, s, s) = 0.$$

Then using the fact that $W(x)$ is an increasing function, we know that

$$W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1}) > W\left(e^{v_1-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}\right).$$

Also note that

$$\frac{c}{2} > \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}.$$

Summing up two parts we have $r_2^* > \tilde{r}$.

Second, we show $r_1^* \leq \tilde{r}$. Note again that the Product Log function $W(\cdot)$ is an increasing function on $[0, +\infty)$, so we have the following.

$$\begin{aligned} \tilde{r} &= W\left(e^{v_1-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}}\right) + 1 + \frac{s}{2} + \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})} \\ &> W\left(e^{v_1-1-s/2-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-s/2-c/2}\right) + 1 + \frac{s}{2} + \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}. \end{aligned}$$

Denote $h(x) = W(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2}) + 1 + s/2 + x$. Then the above inequality reduces to $\tilde{r} > h\left(\frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}\right)$.

Using the expression of the first order derivative of the Product Log function, $W'(x) = \frac{W(x)}{x(1+W(x))}$, we have

$$\begin{aligned} h'(x) &= -W'(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2}) \cdot e^{v_1-1-s/2-x} + 1 \\ &= 1 - \frac{e^{v_1-1-s/2-x} \cdot W(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2})}{(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2})(1 + W(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2}))} \\ &= \frac{(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2}) + e^{v_2-1-s/2-c/2} \cdot W(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2})}{(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2})(1 + W(e^{v_1-1-s/2-x} + e^{v_2-1-s/2-c/2}))} \\ &> 0. \end{aligned}$$

So $h(x)$ is increasing in x . Then we have $h\left(\frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}\right) > h\left(\frac{s}{2}\right)$, which then implies

$$\begin{aligned} \tilde{r} &> W\left(e^{v_1-1-s/2-\frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2-1-s/2-c/2}\right) + 1 + \frac{s}{2} + \frac{ce^{v_2}}{2(e^{v_1} + e^{v_2})} \\ &> W(e^{v_1-1-s} + e^{v_2-1-s/2-c/2}) + 1 + s = r_1^*. \end{aligned}$$

So we have $\tilde{r} > r_1^*$. $\square$

EC.5. Proof of Theorem 5

Proof of Theorem 5. This proof consists of two parts.

Part 1. We prove the sensitivity results for the optimal prices. First, we focus on $\tilde{r}$. Recall that, when $\text{WASTE}(\tilde{r}, \tilde{r}) > \text{WASTE}(r_1^*, r_2^*) > 0$, the optimal fixed price can be expressed by (EC.9), which we re-write below,

$$\tilde{r} = W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}}\right) + 1 + \frac{s}{2} + \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}.$$

Denote

$$\delta = 1 + \frac{s}{2} + \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})},$$

and we know δ is increasing in s . Using δ , we can express $\tilde{r}$ as

$$\tilde{r} = W(e^{v_1-\delta} + e^{v_2-\delta}) + \delta.$$

Then we take the first order derivative,

$$\begin{aligned} \frac{d\tilde{r}}{d\delta} &= -W'(e^{v_1-\delta} + e^{v_2-\delta})(e^{v_1-\delta} + e^{v_2-\delta}) + 1 \\ &= -\frac{W(e^{v_1-\delta} + e^{v_2-\delta})}{1 + W(e^{v_1-\delta} + e^{v_2-\delta})} + 1 \\ &= \frac{1}{1 + W(e^{v_1-\delta} + e^{v_2-\delta})} \\ &> 0. \end{aligned}$$

Since $\tilde{r}$ is increasing in δ and δ is increasing in s , we know $\tilde{r}$ is increasing in s .

Second, we focus on r_1^* . Recall that, when $\text{WASTE}(\tilde{r}, \tilde{r}) > \text{WASTE}(r_1^*, r_2^*) > 0$, the optimal optimal mark-down prices can be expressed by (EC.14), which we re-write below,

$$r_1^* = W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1}) + 1 + s.$$

We take the first order derivative,

$$\begin{aligned} \frac{dr_1^*}{ds} &= -W'(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})(e^{v_1-s-1} + \frac{1}{2}e^{v_2-s/2-c/2-1}) + 1 \\ &= -\frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1 + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})} \cdot \frac{e^{v_1-s-1} + \frac{1}{2}e^{v_2-s/2-c/2-1}}{e^{v_1-s-1} + e^{v_2-s/2-c/2-1}} + 1 \\ &> 0, \end{aligned}$$

where the last inequality holds because both $\frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1 + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}$ and $\frac{e^{v_1-s-1} + \frac{1}{2}e^{v_2-s/2-c/2-1}}{e^{v_1-s-1} + e^{v_2-s/2-c/2-1}}$ are smaller than 1. So r_1^* is increasing in s .

Third, we show that r_2^* is undetermined. We construct two examples. In the first example, $v_1 = 1, v_2 = 0.5, c = 1$, and $s \in (-1, 1)$. So r_2^* is increasing in s . See the left figure in Figure 4. In the second example, $v_1 = 10, v_2 = 9, c = 1$, and $s \in (-1, 1)$. So r_2^* is decreasing in s . See the right figure in Figure 4.

Part 2. We prove the sensitivity results for the wastes. First, we focus on $\text{WASTE}(\tilde{r}, \tilde{r})$. Recall that, when $\text{WASTE}(\tilde{r}, \tilde{r}) > \text{WASTE}(r_1^*, r_2^*) > 0$, the waste under the optimal fixed price can be expressed by (EC.8), which we re-write below,

$$\text{WASTE}(\tilde{r}, \tilde{r}) = q - \lambda \cdot \frac{2 + e^{v_2-v_1}}{1 + e^{v_2-v_1}} \cdot \frac{W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}}\right)}{1 + W\left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}}\right)}.$$

Again using the notation that $\delta = 1 + \frac{s}{2} + \frac{se^{v_1+ce^{v_2}}}{2(e^{v_1}+e^{v_2})}$, we express

$$\text{WASTE}(\tilde{r}, \tilde{r}) = q - \lambda \cdot \frac{2 + e^{v_2-v_1}}{1 + e^{v_2-v_1}} \cdot \frac{W(e^{v_1-\delta} + e^{v_2-\delta})}{1 + W(e^{v_1-\delta} + e^{v_2-\delta})}.$$

Since $W(e^{v_1-\delta} + e^{v_2-\delta})$ is decreasing in δ , $\text{WASTE}(\tilde{r}, \tilde{r})$ is increasing in δ . Furthermore, δ is increasing in s , so $\text{WASTE}(\tilde{r}, \tilde{r})$ is increasing in s .

Next, we focus on $\text{WASTE}(r_1^*, r_2^*)$. Recall that, when $\text{WASTE}(\tilde{r}, \tilde{r}) > \text{WASTE}(r_1^*, r_2^*) > 0$, the waste under the optimal markdown prices can be expressed by (EC.15), which we re-write below,

$$\text{WASTE}(r_1^*, r_2^*) = q - \lambda \cdot \frac{2 + e^{v_2-v_1+s/2-c/2}}{1 + e^{v_2-v_1+s/2-c/2}} \cdot \frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1 + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}.$$

We focus on the two fractions separately. Note that, $e^{v_2-v_1+s/2-c/2}$ is increasing in s , so the first fraction $\frac{2 + e^{v_2-v_1+s/2-c/2}}{1 + e^{v_2-v_1+s/2-c/2}}$ is decreasing in s . On the other hand, note that $W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})$ is decreasing in s , so the second fraction $\frac{W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}{1 + W(e^{v_1-s-1} + e^{v_2-s/2-c/2-1})}$ is decreasing in s . Combining both parts, $\text{WASTE}(r_1^*, r_2^*)$ is increasing in s . $\square$

EC.6. Proof of Theorem 6

Proof of Theorem 6. This proof consists of two parts.

Part 1. We focus on the fixed price policy by maximizing the profit of $\text{PROFIT}(\tilde{r}^J, \tilde{r}^J, \tilde{q}^J)$. We show that the optimal solution $(\tilde{r}^J, \tilde{r}^J, \tilde{q}^J)$ must be in the case such that $\tau(r, r, q) = 1$. To see this, we discuss $\tau(r, r, q) = 0$ and $0 < \tau(r, r, q) < 1$ cases separately.

If $\tau(r, r, q) = 0$, then $q \leq \lambda p_{2|2}(r)$. The objective of the profit maximization problem reduces to

$$\max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} r \cdot q.$$

The objective function of the profit maximization problem is linearly increasing in q . Till q increases to larger than $\lambda p_{2|2}(r)$, this violates $q \leq \lambda p_{2|2}(r)$, thus violating $\tau(r, r, q) = 0$. As a result, as long as q is such that $\tau(r, r, q) = 0$, this cannot be optimal.

If $0 < \tau(r, r, q) < 1$, then we have both $q > \lambda p_{2|2}(r)$ and $q < 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$. Note that

$$\tau(r, r, q) = \frac{q - \lambda p_{2|2}(r)}{2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)},$$

which implies

$$\begin{aligned} & \tau(r, r, q)\lambda p_{2|1,2}(r, r) + (1 - \tau(r, r, q))\lambda p_{2|2}(r) \\ &= \tau(r, r, q)[\lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)] + \lambda p_{2|2}(r) \\ &= \tau(r, r, q)[2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)] - 2\tau(r, r, q)\lambda p_{1|1,2}(r, r) + \lambda p_{2|2}(r) \\ &= q - \lambda p_{2|2}(r) - 2\lambda p_{1|1,2}(r, r)\tau(r, r, q) + \lambda p_{2|2}(r) \\ &= q - 2\lambda\tau(r, r, q)p_{1|1,2}(r, r). \end{aligned}$$

With all above, the objective of the profit maximization problem reduces to

$$\begin{aligned} & \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - c)\tau(r, r, q)\lambda p_{1|1,2}(r, r) + (r - c)[\tau(r, r, q)\lambda p_{2|1,2}(r, r) + (1 - \tau(r, r, q))\lambda p_{2|2}(r)] \\ &= \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - c)\tau(r, r, q)\lambda p_{1|1,2}(r, r) + (r - c)q - (r - c)2\lambda\tau(r, r, q)p_{1|1,2}(r, r) \\ &= \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - c) \cdot [q - \tau(r, r, q)\lambda p_{1|1,2}(r, r)] \end{aligned}$$

where c and λ are constants. Plugging in the expressions of MNL, the objective of the profit maximization problem can be further expressed as

$$\begin{aligned} & \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - c) \cdot [q - \tau(r, r)\lambda p_{1|1,2}(r, r)] \\ &= \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - c) \cdot \left[q - \frac{q - \lambda p_{2|2}(r)}{2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) - \lambda p_{2|2}(r)} \cdot \lambda p_{1|1,2}(r, r) \right] \\ &= \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - c) \cdot \left[q - (q - \lambda \frac{e^{v_2 - r}}{1 + e^{v_2 - r}}) \frac{1 + e^{v_2 - r}}{2 + e^{v_2 - r}} \right] \\ &= \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - c) \cdot \left[q \cdot \frac{1}{2 + e^{v_2 - r}} + \lambda \cdot \frac{e^{v_2 - r}}{2 + e^{v_2 - r}} \right]. \end{aligned}$$

The objective function of the profit maximization problem is linearly increasing in q . Till q increases to larger than $2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$, this violates $q < 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$, thus violating $0 < \tau(r, r, q) < 1$. As a result, as long as q is such that $0 < \tau(r, r, q) < 1$, this cannot be optimal.

Combining the above two cases, we know that the optimal $(\tilde{r}^J, \tilde{r}^J, \tilde{q}^J)$ must satisfy that $\tau(r, r, q) = 1$. Then we have both $q > \lambda p_{2|2}(r)$ and $q \geq 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$.

The objective of the profit maximization problem reduces to

$$\max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} (r - s)\lambda p_{1|1,2}(r, r) + (r - \frac{s}{2} - \frac{c}{2})\lambda p_{2|1,2}(r, r) + \frac{s - c}{2}q,$$

where c and λ are constants.

Note that $2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) \geq \lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) \geq \lambda p_{2|2}(r)$. And note that $\lambda > 0$. So it suffices to ensure that $q \geq 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r)$. Plugging in the expressions of MNL, the profit maximization problem that we solve reduces to

$$\begin{aligned} \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} & \frac{\lambda(r - s)e^{v_1 - r}}{1 + e^{v_1 - r} + e^{v_2 - r}} + \frac{\lambda(r - s/2 - c/2)e^{v_2 - r}}{1 + e^{v_1 - r} + e^{v_2 - r}} + \frac{s - c}{2} \cdot q \\ \text{s.t.} & \quad 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) \leq q. \end{aligned}$$

Note that the objective function is linearly decreasing in q . So at the optimal solution $(\tilde{r}^J, \tilde{r}^J, \tilde{q}^J)$, it must satisfy that

$$\tilde{q}^J = 2\lambda p_{1|1,2}(\tilde{r}^J, \tilde{r}^J) + \lambda p_{2|1,2}(\tilde{r}^J, \tilde{r}^J). \quad (\text{EC.18})$$

This is the optimality condition of the joint inventory and pricing decision under the fixed price policy.

Under the optimality condition (EC.18), we have $\text{WASTE}(\tilde{r}^J, \tilde{r}^J, \tilde{q}^J) = 0$.

Part 2. We focus on the markdown policy by maximizing the profit of $\text{PROFIT}(r_1^{J*}, r_2^{J*}, q^{J*})$. We show that the optimal solution $(r_1^{J*}, r_2^{J*}, q^{J*})$ must be in the case such that $\tau(r_1, r_2, q) = 1$. To see this, we discuss $\tau(r_1, r_2, q) = 0$ and $0 < \tau(r_1, r_2, q) < 1$ cases separately.

If $\tau(r_1, r_2, q) = 0$, then $q \leq \lambda p_{2|2}(r_2)$. The objective function of the profit maximization problem reduces to

$$\max_{r_2 \in \mathbb{R}^+} r_2 \cdot q.$$

The objective function of the profit maximization problem is linearly increasing in q . Till q increases to larger than $\lambda p_{2|2}(r_2)$, this violates $\tau(r_1, r_2, q) = 0$. As a result, as long as q is such that $\tau(r_1, r_2, q) = 0$, this cannot be optimal.

If $0 < \tau(r_1, r_2, q) < 1$, we have both $q > \lambda p_{2|2}(r_2)$ and $q < 2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2)$. Note that

$$\tau(r_1, r_2, q) = \frac{q - \lambda p_{2|2}(r_2)}{2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) - \lambda p_{2|2}(r_2)},$$

which implies

$$\begin{aligned}
& \tau(r_1, r_2, q)\lambda p_{2|1,2}(r_1, r_2) + (1 - \tau(r_1, r_2, q))\lambda p_{2|2}(r_2) \\
&= \tau(r_1, r_2, q)(\lambda p_{2|1,2}(r_1, r_2) - \lambda p_{2|2}(r_2)) + \lambda p_{2|2}(r_2) \\
&= \tau(r_1, r_2, q)(2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) - \lambda p_{2|2}(r_2)) - 2\tau(r_1, r_2, q)\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|2}(r_2) \\
&= q - \lambda p_{2|2}(r_2) - 2\lambda p_{1|1,2}(r_1, r_2)\tau(r_1, r_2, q) + \lambda p_{2|2}(r_2) \\
&= q - 2\lambda\tau(r_1, r_2, q)p_{1|1,2}(r_1, r_2).
\end{aligned}$$

With all above, the profit maximization problem reduces to

$$\begin{aligned}
& \max_{r_1, r_2, q \in \mathbb{R}^+} (r_1 - c)\tau(r_1, r_2, q)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - c) [\tau(r_1, r_2, q)\lambda p_{2|1,2}(r_1, r_2) + (1 - \tau(r_1, r_2, q))\lambda p_{2|2}(r_2)] \\
&= \max_{r_1, r_2, q \in \mathbb{R}^+} (r_1 - c)\tau(r_1, r_2, q)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - c)q - (r_2 - c)2\lambda\tau(r_1, r_2, q)p_{1|1,2}(r_1, r_2) \\
&= \max_{r_1, r_2, q \in \mathbb{R}^+} \tau(r_1, r_2, q)\lambda p_{1|1,2}(r_1, r_2)(r_1 - 2r_2 + c) + (r_2 - c)q
\end{aligned}$$

where c and λ are constants.

Note that, by plugging in the expressions of the choice probabilities,

$$\begin{aligned}
& \tau(r_1, r_2, q)\lambda p_{1|1,2}(r_1, r_2)(r_1 - 2r_2 + c) + (r_2 - c)q \\
&= (q - \lambda p_{2|2}(r_2)) \frac{p_{1|1,2}(r_1, r_2)}{2p_{1|1,2}(r_1, r_2) + p_{2|1,2}(r_1, r_2) - p_{2|2}(r_2)} (r_1 - 2r_2 + c) + (r_2 - c)q \\
&= (q - \lambda p_{2|2}(r_2)) \frac{1 + e^{v_2 - r_2}}{2 + e^{v_2 - r_2}} (r_1 - 2r_2 + c) + (r_2 - c)q.
\end{aligned}$$

Because $q > \lambda p_{2|2}(r_2)$, we know that the objective function of the profit maximization problem is monotone increasing in r_1 . But as r_1 increases, the quantity

$$2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) = \lambda \cdot \frac{2e^{v_1 - r_1} + e^{v_2 - r_2}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}}$$

decreases. Till $2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2)$ decreases to q , this violates $\tau(r_1, r_2, q) < 1$. As a result, as long as r_1, r_2 , and q makes $\tau(r_1, r_2, q) < 1$, this cannot be optimal.

From above, we know the optimal solution $(r_1^{j*}, r_2^{j*}, q^{j*})$ must be in the case such that $\tau(r_1, r_2, q) = 1$.

When $\tau(r_1, r_2, q) = 1$, we have both $q > \lambda p_{2|2}(r_2)$ and $q \geq 2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2)$.

Then the profit maximization problem reduces to

$$\begin{aligned}
& \max_{r_1, r_2 \in \mathbb{R}^+} (r_1 - c)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - c)\lambda p_{2|1,2} + \frac{s - c}{2}(q - 2\lambda p_{1|1,2}(r_1, r_2) - \lambda p_{2|1,2}(r_1, r_2)) \\
&= \max_{r_1, r_2 \in \mathbb{R}^+} (r_1 - s)\lambda p_{1|1,2}(r_1, r_2) + (r_2 - \frac{s}{2} - \frac{c}{2})\lambda p_{2|1,2}(r_1, r_2) + \frac{s - c}{2}q,
\end{aligned}$$

where c and λ are constants. Plugging in the expressions of MNL we re-write the profit maximization problem as:

$$\begin{aligned} \max_{r_1, r_2 \in \mathbb{R}^+, q \in \mathbb{R}^+} & \frac{\lambda(r_1 - s)e^{v_1 - r_1}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}} + \frac{\lambda(r_2 - s/2 - c/2)e^{v_2 - r_2}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}} + \frac{s - c}{2}q \\ \text{s.t.} & \quad 2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) \leq q. \end{aligned}$$

Note that the objective function is linearly decreasing in q . So at the optimal solution $(r_1^{J*}, r_2^{J*}, q^{J*})$, it must satisfy that

$$q^{J*} = 2\lambda p_{1|1,2}(r_1^{J*}, r_2^{J*}) + \lambda p_{2|1,2}(r_1^{J*}, r_2^{J*}). \quad (\text{EC.19})$$

This is the optimality condition of the joint inventory and pricing decision under the markdown policy.

Under the optimality condition (EC.19), we have $\text{WASTE}(r_1^{J*}, r_2^{J*}, q^{J*}) = 0$. $\square$

EC.7. Proof of Theorem 8

Proof of Theorem 8. In the proof of Lemma EC.5, we have shown that if the following condition holds

$$\frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}{1 + W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)} < \frac{q}{\lambda},$$

then

$$\tilde{r} = W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right) + 1 + \frac{s}{2} + \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}.$$

follows from (EC.9). The proof of Theorem 3 suggests that when $\text{WASTE}(\tilde{r}, \tilde{r}, q) > \text{WASTE}(r_1^*, r_2^*, q) > 0$, the condition

$$\frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}{1 + W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)} < \frac{q}{\lambda}$$

holds. Furthermore, when $r_1 = r_2 = \tilde{r}$, the proof of Lemma EC.5 suggests that the optimal profit can be expressed as

$$\text{PROFIT}(\tilde{r}, \tilde{r}, q) = \frac{\lambda(\tilde{r} - s)e^{v_1 - \tilde{r}}}{1 + e^{v_1 - \tilde{r}} + e^{v_2 - \tilde{r}}} + \frac{\lambda(\tilde{r} - s/2 - c/2)e^{v_2 - \tilde{r}}}{1 + e^{v_1 - \tilde{r}} + e^{v_2 - \tilde{r}}} + \frac{s - c}{2} \cdot q.$$

From here, notice that $s < c$ so the profit function $\text{PROFIT}(\tilde{r}, \tilde{r}, q)$ is linearly decreasing in q . As a result, when $r_1 = r_2 = \tilde{r}$ remains fixed, the profit maximizing order-up-to level q should be such that

$$\check{q}(\tilde{r}) = \lambda \cdot \frac{2 + e^{v_2 - v_1}}{1 + e^{v_2 - v_1}} \cdot \frac{W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}{1 + W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)}.$$

This immediately suggests that

$$\text{WASTE}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) = 0 < \text{WASTE}(r_1^*, r_2^*, q).$$

Next, we compare $\text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r}))$ and $\text{PROFIT}(r_1^*, r_2^*, q)$. We wish to find problem instances $v_1, v_2, s, c, q, \lambda$ such that

$$\text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) < \text{PROFIT}(r_1^*, r_2^*, q),$$

and problem instances $v_1, v_2, s, c, q, \lambda$ such that

$$\text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) > \text{PROFIT}(r_1^*, r_2^*, q).$$

See the two parts below.

Part 1. Consider the case when q is given to be $q = 2p_{1|1,2}(r_1^*, r_2^*) + p_{2|1,2}(r_1^*, r_2^*)$. This is the “low-waste” inventory regime. We wish to show that under this low-waste inventory regime, there exist problem instances such that pricing decisions are more important than inventory decisions.

Specifically, we wish to show that there exists $c_0 = s + \varepsilon > s$ such that when $c \in (s, c_0)$,

$$\text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) - \text{PROFIT}(r_1^*, r_2^*, q) < 0, \quad (\text{EC.20})$$

At $r_1 = r_2 = \tilde{r}$, we choose the optimal inventory level $\check{q}(\tilde{r}) = 2p_{1|1,2}(\tilde{r}, \tilde{r}) + p_{2|1,2}(\tilde{r}, \tilde{r})$. Then we have

$$\text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) = (\tilde{r} - c) \frac{e^{v_1 - \tilde{r}} + e^{v_2 - \tilde{r}}}{1 + e^{v_1 - \tilde{r}} + e^{v_2 - \tilde{r}}}$$

Denote $\tilde{W} = W \left(e^{v_1 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} + e^{v_2 - 1 - \frac{s}{2} - \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})}} \right)$. Putting this into the expression above, we have

$$\begin{aligned} \text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) &= \left(\tilde{W} + 1 + \frac{s}{2} + \frac{se^{v_1} + ce^{v_2}}{2(e^{v_1} + e^{v_2})} - c \right) \cdot \frac{\tilde{W}}{1 + \tilde{W}} \\ &= \frac{\tilde{W}}{1 + \tilde{W}} \cdot \frac{(\tilde{W} + 1 + s - c)e^{v_1} + (\tilde{W} + 1 + \frac{s}{2} - \frac{c}{2})e^{v_2}}{e^{v_1} + e^{v_2}} := \tilde{h}(v_1, v_2, s, c), \end{aligned}$$

where the first equality follows (EC.9).

On the other hand,

$$\text{PROFIT}(r_1^*, r_2^*, q) = \frac{(r_1^* - c)e^{v_1 - r_1^*} + (r_2^* - c)e^{v_2 - r_2^*}}{1 + e^{v_1 - r_1^*} + e^{v_2 - r_2^*}}$$

Denote $W^* = W \left(e^{v_1 - s - 1} + e^{v_2 - s/2 - c/2 - 1} \right)$. Putting this into the expression above, we have

$$\begin{aligned} \text{PROFIT}(r_1^*, r_2^*, q) &= \frac{(W^* + 1 + s - c)e^{v_1 - 1 - s - W^*} + (W^* + 1 + \frac{s}{2} - \frac{c}{2})e^{v_2 - 1 - \frac{s}{2} - \frac{c}{2} - W^*}}{1 + W^*} \\ &= \frac{W^*}{1 + W^*} \cdot \frac{(W^* + 1 + s - c)e^{v_1} + (W^* + 1 + \frac{s}{2} - \frac{c}{2})e^{v_2 + \frac{s}{2} - \frac{c}{2}}}{e^{v_1} + e^{v_2 + \frac{s}{2} - \frac{c}{2}}} := h^*(v_1, v_2, s, c), \end{aligned}$$

where the first equality follows (EC.14).

Now we study the behavior of $\tilde{h}(v_1, v_2, s, c)$ and $h^*(v_1, v_2, s, c)$ when c is close to s . First, note that when $c = s$, we have $\tilde{W} = W^*$ so $\tilde{h}(v_1, v_2, s, c) = h^*(v_1, v_2, s, c)$. Next, we compare the first order derivatives of $\frac{\partial \tilde{h}(v_1, v_2, s, c)}{\partial c} \Big|_{c=s}$ and $\frac{\partial h^*(v_1, v_2, s, c)}{\partial c} \Big|_{c=s}$ at $c = s$.

Note that for the Product Log function $W(\gamma)$, $W'(\gamma) = \frac{W(\gamma)}{(1+W(\gamma))\gamma}$. Denote the first order derivative $\frac{\partial \tilde{W}(v_1, v_2, s, c)}{\partial c}$ as $(\tilde{W})'$. Then $(\tilde{W})'$ can be expressed as

$$(\tilde{W})' = \frac{\tilde{W}(e^{v_1} + e^{v_2})e^{-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2e^{v_1}+2e^{v_2}}} \cdot \left(-\frac{e^{v_2}}{2e^{v_1}+2e^{v_2}}\right)}{(1+\tilde{W})(e^{v_1} + e^{v_2})e^{-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2e^{v_1}+2e^{v_2}}}} = \frac{-\tilde{W}e^{v_2}}{2(1+\tilde{W})(e^{v_1} + e^{v_2})}$$

Using $(\tilde{W})'$ as a succinct notation, we have

$$\begin{aligned} \frac{\partial \tilde{h}(v_1, v_2, s, c)}{\partial c} &= \frac{1}{(1+\tilde{W})^2} \cdot (\tilde{W})' \cdot \frac{(\tilde{W}+1+s-c)e^{v_1} + (\tilde{W}+1+\frac{s}{2}-\frac{c}{2})e^{v_2}}{e^{v_1} + e^{v_2}} \\ &\quad + \frac{\tilde{W}}{1+\tilde{W}} \cdot \left(\frac{e^{v_1}}{e^{v_1} + e^{v_2}} \left((\tilde{W})' - 1 \right) + \frac{e^{v_2}}{e^{v_1} + e^{v_2}} \left((\tilde{W})' - \frac{1}{2} \right) \right). \end{aligned}$$

Furthermore, we evaluate the first order derivative at $c = s$,

$$\begin{aligned} \frac{\partial \tilde{h}(v_1, v_2, s, c)}{\partial c} \Big|_{c=s} &= \frac{1}{1+\tilde{W}} \cdot \frac{-\tilde{W}e^{v_2}}{2(1+\tilde{W})(e^{v_1} + e^{v_2})} \\ &\quad + \frac{\tilde{W}}{1+\tilde{W}} \cdot \left(\frac{-\tilde{W}e^{v_2}}{2(1+\tilde{W})(e^{v_1} + e^{v_2})} - 1 \right) \\ &\quad + \frac{\tilde{W}}{1+\tilde{W}} \cdot \frac{e^{v_2}}{2(e^{v_1} + e^{v_2})}. \end{aligned} \tag{EC.21}$$

Denote the first order derivative $\frac{\partial W^*(v_1, v_2, s, c)}{\partial c}$ as $(W^*)'$. Then $(W^*)'$ can be expressed as

$$(W^*)' = \frac{-W^*e^{v_2-\frac{s}{2}-\frac{c}{2}-1}}{2(1+W^*)(e^{v_1-s-1} + e^{v_2-\frac{s}{2}-\frac{c}{2}-1})}$$

Using $(W^*)'$ as a succinct notation, we have

$$\begin{aligned} &\frac{\partial h^*(v_1, v_2, s, c)}{\partial c} \\ &= \frac{1}{(1+W^*)^2} \cdot (W^*)' \cdot \frac{(W^*+1+s-c)e^{v_1} + (W^*+1+\frac{s}{2}-\frac{c}{2})e^{v_2+\frac{s}{2}-\frac{c}{2}}}{e^{v_1} + e^{v_2+\frac{s}{2}-\frac{c}{2}}} \\ &\quad + \frac{W^*}{1+W^*} \cdot \frac{1}{e^{v_1} + e^{v_2+\frac{s}{2}-\frac{c}{2}}} \left(\left((W^*)' - 1 \right) e^{v_1} + \left((W^*)' - \frac{1}{2} \right) e^{v_2+\frac{s}{2}-\frac{c}{2}} - \frac{1}{2} \left((W^*)' + 1 + \frac{s}{2} - \frac{c}{2} \right) e^{v_2+\frac{s}{2}-\frac{c}{2}} \right) \\ &\quad + \frac{W^*}{1+W^*} \cdot \frac{1}{(e^{v_1} + e^{v_2+\frac{s}{2}-\frac{c}{2}})^2} \left((W^*+1+s-c)e^{v_1} + (W^*+1+\frac{s}{2}-\frac{c}{2})e^{v_2+\frac{s}{2}-\frac{c}{2}} \right) \frac{1}{2} e^{v_2+\frac{s}{2}-\frac{c}{2}}. \end{aligned}$$

Furthermore, we evaluate the first order derivative at $c = s$,

$$\frac{\partial h^*(v_1, v_2, s, c)}{\partial c} \Big|_{c=s} = \frac{1}{(1+W^*)^2} \cdot \frac{-W^*e^{v_2-s-1}}{2(1+W^*)(e^{v_1-s-1} + e^{v_2-s-1})} \cdot \frac{(W^*+1)e^{v_1} + (W^*+1)e^{v_2}}{e^{v_1} + e^{v_2}}$$

$$\begin{aligned}
& + \frac{W^*}{1+W^*} \cdot \frac{1}{e^{v_1}+e^{v_2}} \left(\frac{-W^*e^{v_2-s-1}}{2(1+W^*)(e^{v_1-s-1}+e^{v_2-s-1})} - 1 \right) (e^{v_1}+e^{v_2}) \\
& + \frac{W^*}{1+W^*} \cdot \frac{1}{e^{v_1}+e^{v_2}} \left(\frac{1}{2}e^{v_2} - \frac{1}{2} \left(\frac{-W^*e^{v_2-s-1}}{2(1+W^*)(e^{v_1-s-1}+e^{v_2-s-1})} + 1 \right) e^{v_2} \right) \\
& + \frac{W^*}{1+W^*} \cdot \frac{1}{(e^{v_1}+e^{v_2})^2} \left((W^*+1)e^{v_1} + (W^*+1)e^{v_2} \right) \frac{1}{2}e^{v_2} \\
& = \frac{1}{1+W^*} \cdot \frac{-W^*e^{v_2}}{2(1+W^*)(e^{v_1}+e^{v_2})} \\
& + \frac{W^*}{1+W^*} \cdot \left(\frac{-W^*e^{v_2}}{2(1+W^*)(e^{v_1}+e^{v_2})} - 1 \right) \\
& + \frac{W^*}{1+W^*} \cdot \frac{1}{e^{v_1}+e^{v_2}} \frac{W^*e^{v_2}}{4(1+W^*)(e^{v_1}+e^{v_2})} e^{v_2} \\
& + \frac{W^*}{1+W^*} \cdot \frac{1}{e^{v_1}+e^{v_2}} (W^*+1) \frac{1}{2}e^{v_2}. \tag{EC.22}
\end{aligned}$$

Recall that $\tilde{W} = W^*$. Combining (EC.21) and (EC.22) we have

$$\begin{aligned}
& \left. \frac{\partial \tilde{h}(v_1, v_2, s, c) - h^*(v_1, v_2, s, c)}{\partial c} \right|_{c=s} \\
& = -\frac{W^*}{1+W^*} \cdot \frac{1}{e^{v_1}+e^{v_2}} \frac{W^*e^{v_2}}{4(1+W^*)(e^{v_1}+e^{v_2})} e^{v_2} - \frac{W^*}{1+W^*} \cdot \frac{1}{e^{v_1}+e^{v_2}} W^* \frac{1}{2}e^{v_2} < 0.
\end{aligned}$$

Because $\tilde{h}(v_1, v_2, s, s) - h^*(v_1, v_2, s, s) = 0$, this means that there exists $c_0 = s + \varepsilon > s$ such that when $c \in (s, c_0)$, we always have $\tilde{h}(v_1, v_2, s, c) - h^*(v_1, v_2, s, c) < 0$. This proves (EC.20).

Part 2. Consider the case when q is given by $\text{PROFIT}(r_1^*, r_2^*, q) = 0$. This is the “high-waste” inventory regime. We wish to show that under this high-waste inventory regime, there exist problem instances such that inventory decisions are more important than pricing decisions.

Specifically, we wish to show that when $c \in (s, s+1)$,

$$\text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) - \text{PROFIT}(r_1^*, r_2^*, q) > 0, \tag{EC.23}$$

This is not hard to show. Denote $\tilde{W} = W \left(e^{v_1-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} + e^{v_2-1-\frac{s}{2}-\frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})}} \right)$. Using this expression, we have

$$\begin{aligned}
\text{PROFIT}(\tilde{r}, \tilde{r}, \check{q}(\tilde{r})) & = \left(\tilde{W} + 1 + \frac{s}{2} + \frac{se^{v_1}+ce^{v_2}}{2(e^{v_1}+e^{v_2})} - c \right) \cdot \frac{\tilde{W}}{1+\tilde{W}} \\
& = \frac{\tilde{W}}{1+\tilde{W}} \cdot \frac{(\tilde{W}+1+s-c)e^{v_1} + (\tilde{W}+1+\frac{s}{2}-\frac{c}{2})e^{v_2}}{e^{v_1}+e^{v_2}} > \frac{(\tilde{W})^2}{1+\tilde{W}} > 0,
\end{aligned}$$

where the equality follows (EC.9); the first inequality is because $c < s+1$. This proves (EC.23). $\square$

EC.8. Proof of Proposition 7

Proof of Proposition 7. This proof consists of two parts.

Part 1. We focus on the fixed price policy by maximizing the profit of $\text{PROFIT}(\tilde{r}^J, \tilde{r}^J, \tilde{q}^J)$. As we have shown in Theorem 6, the profit maximization reduces to

$$(\tilde{r}^J, \tilde{q}^J) = \arg \max_{r \in \mathbb{R}^+, q \in \mathbb{R}^+} \frac{\lambda(r-s)e^{v_1-r}}{1+e^{v_1-r}+e^{v_2-r}} + \frac{\lambda(r-s/2-c/2)e^{v_2-r}}{1+e^{v_1-r}+e^{v_2-r}} + \frac{s-c}{2} \cdot q$$

$$s.t. \ 2\lambda p_{1|1,2}(r, r) + \lambda p_{2|1,2}(r, r) \leq q,$$

and we have the optimality condition

$$\tilde{q}^J = 2\lambda p_{1|1,2}(\tilde{r}^J, \tilde{r}^J) + \lambda p_{2|1,2}(\tilde{r}^J, \tilde{r}^J).$$

Putting the constraint into the profit maximization problem we have the following unconstrained maximization problem

$$\max_{r \in \mathbb{R}^+} \frac{\lambda(r-c)(e^{v_1-r} + e^{v_2-r})}{1+e^{v_1-r}+e^{v_2-r}}. \quad (\text{EC.24})$$

Denote $\check{f}(r) = \frac{(r-c)(e^{v_1-r} + e^{v_2-r})}{1+e^{v_1-r}+e^{v_2-r}} = (r-c) - \frac{r-c}{1+e^{v_1-r}+e^{v_2-r}}$. Then we have

$$\check{f}'(r) = 1 - \frac{(1+e^{v_1-r} + e^{v_2-r}) + (r-c)(e^{v_1-r} + e^{v_2-r})}{(1+e^{v_1-r} + e^{v_2-r})^2}.$$

Setting $\check{f}'(r) = 0$ we have

$$e^{v_1-r} + e^{v_2-r} = r - c - 1,$$

thus,

$$\tilde{r}^J = W(e^{v_1-c-1} + e^{v_2-c-1}) + c + 1.$$

To conclude that this is the optimal solution, first observe that this is a local maximum. Then, $\check{f}(0) < 0$ and $\lim_{r \rightarrow +\infty} \check{f}(r) = 0$. Since $\check{f}(r)$ is a univariate function with only one single local maximum point that satisfies $\check{f}(\tilde{r}) > 0$, we have that $\tilde{r}^J = W(e^{v_1-c-1} + e^{v_2-c-1}) + c + 1$ is an optimal solution.

Part 2. We focus on the fixed price policy by maximizing the profit of $\text{PROFIT}(r_1^{J*}, r_2^{J*}, q^{J*})$. As we have shown in Theorem 6, the profit maximization reduces to

$$(r_1^{J*}, r_2^{J*}, q^{J*}) = \arg \max_{r_1, r_2 \in \mathbb{R}^+, q \in \mathbb{R}^+} \frac{\lambda(r_1-s)e^{v_1-r_1}}{1+e^{v_1-r_1}+e^{v_2-r_2}} + \frac{\lambda(r_2-s/2-c/2)e^{v_2-r_2}}{1+e^{v_1-r_1}+e^{v_2-r_2}} + \frac{s-c}{2} q$$

$$s.t. \ 2\lambda p_{1|1,2}(r_1, r_2) + \lambda p_{2|1,2}(r_1, r_2) \leq q,$$

and we have the optimality condition

$$q^{J*} = 2\lambda p_{1|1,2}(r_1^{J*}, r_2^{J*}) + \lambda p_{2|1,2}(r_1^{J*}, r_2^{J*}).$$

Putting the optimality condition into the profit maximization problem we have the following unconstrained maximization problem

$$\max_{r_1, r_2 \in \mathbb{R}^+} \frac{\lambda(r_1 - c)e^{v_1 - r_1}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}} + \frac{\lambda(r_2 - c)e^{v_2 - r_2}}{1 + e^{v_1 - r_1} + e^{v_2 - r_2}}.$$

Due to the constant markup property (see, e.g., Aydin and Ryan (2000, Proposition 5.1), Maddah and Bish (2007, Lemma 7) Dong et al. (2009, Corollary 2), Akçay et al. (2010, Theorem 6), Wang (2012, Proposition 1)), the optimal prices should satisfy $r_1^{J*} = r_2^{J*} = r^{J*}$. Replacing r_1^{J*} and r_2^{J*} with r^{J*} we re-write the unconstrained maximization problem as

$$r^{J*} = \arg \max_{r \in \mathbb{R}^+} \frac{\lambda(r - c)(e^{v_1 - r} + e^{v_2 - r})}{1 + e^{v_1 - r} + e^{v_2 - r}}.$$

This is exactly the same problem as (EC.24). As a result, $r_1^{J*} = r_2^{J*} = \tilde{r}^J$ and, thus, $q^{J*} = \tilde{q}^J$. $\square$